

The Mathematics and Physics of *Coupling-Gated Cortical Coherence*

A Comprehensive Plain-English Primer

Companion document to the manuscript and its Supplemental Material

Abstract

This primer explains, from the ground up, every piece of advanced mathematics and physics used in the manuscript *Coupling-gated cortical coherence* and its Supplemental Material. It is written for readers with a background in cognitive neuroscience, psychology, machine learning or philosophy of mind who want to follow the argument without reading Lean 4 source or chasing down each cited result. Its parts cover topology and symmetry breaking, information theory and Landauer’s principle, predictive thermodynamics, operator theory on Hilbert spaces, the deterministic and stochastic Kuramoto model, the mean-field synchronization transition and its critical exponent, finite-size and finite-rate corrections, discretization and continuum limits, sheaf theory, topological phase obstructions, metric fixed points, measure-theoretic comparisons of coupling architectures, the formal methodology of the Lean development, circular statistics and the EEG analysis, dimensional analysis, and the interpretive commitment the whole construction is asked to support. Each topic follows one rubric: what the object is, why the framework needs it, what is actually proved, and what is not. No proof is reproduced in full; two are sketched because their structure is unusual. The aim is literacy, not mastery.

This document is a companion, not part of the publication. Where it differs from the manuscript or the Supplemental Material, those are authoritative.

Contents

Part I. Orientation	5
1 What the argument is, and where the mathematics sits	5
1.1 The nine nodes and the eight gaps	5
1.2 How to read this primer	6
2 Notation	6
Part II. Topology, symmetry breaking, and boundaries	7
3 Spontaneous symmetry breaking	7
4 Topological defects: the Kibble classification	8
Part III. Information, entropy, and the cost of forgetting	9
5 Finite capacity	9
6 Landauer’s principle	9
6.1 The Norton–Shenker distinction	10
Part IV. Predictive thermodynamics	10

7	Mutual information and the data processing inequality	10
8	The Still–Sivak–Bell–Crooks bound	11
8.1	Deriving Still’s bound from Landauer’s	12
9	What does not follow: the free-energy reading	12
	Part V. Operators, gradients, and the squared-drift objective	13
10	The squared-drift functional and its gradient	13
	Part VI. The Kuramoto model: dynamics	14
11	The model	14
12	The Lyapunov potential and gradient descent	14
13	The rotating frame	15
14	Do the trajectories exist? Picard–Lindelöf, globally	15
15	Where the trajectories go: Barbalat and Łojasiewicz	16
15.1	The motion stops	16
15.2	Where it stops	16
15.3	The compactness objection, met without an invariant region	17
16	The open gap: propagation of chaos	17
	Part VII. The synchronization transition	18
17	The von Mises stationary density	18
18	Why the threshold is $K_c = 2D$	18
19	The fold-and-cross proof of uniqueness	20
	Part VIII. The critical exponent and the square-root onset	21
20	The exponent $\beta = 1/2$	21
21	From exponent to prediction	22
	Part IX. Finite populations, finite ramps, and what simulation adds	22
22	Euler–Maruyama integration	23
23	Ramp speed sets the apparent onset exponent	23
24	Spatial coupling and geometric frustration	24
25	Squared-drift plasticity does not learn structure	24
	Part X. From discrete to continuum	25
26	Building a coupling matrix from a field	25

27 The negative result: coarse-graining produces no kernel	26
Part XI. Sheaves, gluing, and the unity of a state	27
28 What a sheaf is	27
29 Compatibility is not phase agreement	27
30 Where uniqueness comes from	28
Part XII. Topological obstructions in the phase field	29
31 Twisted gluing, and why it fails to add content	29
32 Phase singularities and winding numbers	29
Part XIII. Metric spaces, contractions, and the reflexive self	30
33 Banach's fixed-point theorem	30
34 The metric on global sections, and why the obvious ones fail	31
34.1 The germ-measure dictionary	31
35 The avatar: making the self-model run through a region	31
Part XIV. Coupling support, atomless measures, and hardware	32
36 Resource-matched coupling reallocation	32
37 The continuum/discrete separation	33
Part XV. Formal method: why the development declares no axioms	33
38 What formalization is claimed to add	33
39 Why a global axiom is the wrong shape for a physical postulate	34
40 Witnesses, and what inhabitation does and does not prove	34
41 The chain theorem	35
Part XVI. Circular statistics and the empirical test	35
42 Doing statistics on angles	35
43 Getting the phases: the analytic signal	35
44 The parameter-free prediction, and the trap in testing it	36
45 What the EEG analysis found	36
46 The awakening experiment	37
Part XVII. Dimensional analysis and the calibration problem	38
47 Why the units do not match	38

48 Why γ is not recoverable	38
49 The empirical case for the field, and its critics	39
Part XVIII. What the mathematics is asked to mean	40
50 The commitment	40
51 Three arguments, and what kind of arguments they are	40
52 The costs	41
53 What would count against it, and the observable that is missing	41
53.1 Three components	41
53.2 Three ways it fails silently	42
54 Relation to other accounts	42
Appendices	42
A Master table: concept, role, and status	42
B Glossary	44
C Where to look next	46

Part I. Orientation

1 What the argument is, and where the mathematics sits

The manuscript makes three separable claims, and it is worth having them apart before any mathematics arrives, because almost every scope caveat in this primer is a statement about which of the three a given theorem touches.

1. **A physical hypothesis.** Cortical regions are coupled partly through the endogenous extracellular electromagnetic field; that coupling is slowly gated; and recovery of coherence after awakening is driven by the coupling parameter drifting up through a synchronization threshold. This is testable with current methods, and most of the manuscript is about testing it.
2. **A formal composition.** Nine mathematical results—finite information capacity, a symmetry-breaking boundary, Landauer’s heat cost, a predictive-dissipation bound, a continuum coupling limit, a physical field realizing it, a unique coherent order parameter, a unique glued global section, and a unique reflexive fixed point—are chained into a single conditional theorem, machine-checked in Lean 4. The chain consumes *eight* implications between those nine nodes, and no theorem in the development supplies them. Naming them is the point of the exercise.
3. **An interpretive commitment.** The unique glued global section is identified with the unity of a conscious episode, and the fixed point of the self-prediction map on it with a minimal self. This is argued for, not proved, and the manuscript says so.

The three can fail independently. The cortical hypothesis can be wrong while the composition stands; the composition can stand while the identification is wrong. Keeping them separate is what makes each assessable, and it is also why this primer repeatedly ends a section by saying what a result does *not* establish. Those are not disclaimers bolted on afterwards—in this framework they are load-bearing, because the whole methodological claim is that formalization makes an argument’s assumptions impossible to lose.

1.1 The nine nodes and the eight gaps

The composition runs, in order:

- n1** a finite phase space has bounded information capacity
- n2** a broken symmetry forces a boundary, where the field leaves its vacuum manifold
- n3** the boundary erases, and erasure costs heat
- n4** that dissipation bounds nonpredictive memory
- n5** discrete couplings coarse-grain to a continuum limit
- n6** a specified physical field realizes that limiting coupling
- n7** above threshold the field carries a unique coherent order parameter
- n8** compatible local states glue to a unique global section
- n9** that section is the unique fixed point of a self-prediction map

Each node is established or cited. Each *arrow* between consecutive nodes is an assumption, labelled **E12** through **E89**, and the manuscript classifies each one by kind:

Edge	Kind	What it asks for
E12	independent physical premise	a finite-capacity system realizes a field leaving its vacuum manifold
E23	formalization gap	the boundary is a finite register whose update strands states
E34	formalization gap	erasure supplies predictive structure at the same thermal scale
E45	modelling assumption	the dissipation bound places couplings in a convergent regime
E56	physical commitment	the limiting coupling is realized by a specified physical field
E67	physical commitment	that field operates above the synchronization threshold
E78	formalization gap	coherence drives a <i>given</i> cover to its relaxed equilibrium
E89	modelling assumption	a contracting map fixes the section glued on that cover

Two features of that table matter. First, the two *physical commitments* (E56, E67) are the empirically testable core, and they are bracketed by formalization gaps on either side—so the chain’s empirical content sits between links of a different kind. Second, E12 is drawn as a premise rather than a gap because n_1 and n_2 are independent roots: finite capacity does not imply a symmetry-breaking boundary, and the composition merely accepts an implication between them.

1.2 How to read this primer

Parts II–XVI are ordered roughly by the chain, then by the methodological and empirical material that surrounds it. Every subsection uses the same four headings—**What it is**, **Why the framework needs it**, **The statement**, **What it does not give**—plus an **In Lean** note where a formal identifier carries the claim. You can read any subsection on its own; forward references are marked.

Part XIV, on formal methodology, is the one part with no physics in it. It is included because the manuscript’s fourth contribution is a claim about *method*—that a machine-checked composition makes an assumption list non-negotiable—and that claim rests on a specific technical point about how physical postulates are encoded. Readers who want the physics only may skip it; readers who want to know why the development declares no axioms should not.

2 Notation

Symbol	Meaning
θ_i	phase of oscillator i , an angle on the circle $[-\pi, \pi)$
ω_i	natural (intrinsic) frequency of oscillator i
K, K_{ij}	coupling strength; the coupling matrix. Inverse-time units
$K(x, y)$	continuum coupling kernel on a spatial substrate
D	phase diffusion coefficient (noise strength). Inverse-time units
K_c	critical coupling, $= 2D$ in this model
W_i	independent Wiener processes (Brownian motion)
r, r_N	order parameter: circular resultant length, in $[0, 1]$
ψ	collective phase (the direction of the mean resultant)
a	von Mises concentration parameter, $a = Kr/D$
I_n	modified Bessel function of the first kind, order n
$R(a)$	the Bessel ratio $I_1(a)/I_0(a)$
$E(a)$	$\mathbb{E}_a[\sin^2 \theta]$ under the von Mises law at concentration a
β	critical exponent of the order parameter; $= 1/2$ here
$\rho(\theta)$	stationary phase density
$V(\theta)$	Kuramoto Lyapunov potential
σ	squared-drift functional (<i>not</i> , in general, entropy production)
W_{diss}	dissipated work

Symbol	Meaning
$I(X; Y)$	mutual information
$D_{KL}(P Q)$	Kullback–Leibler divergence
S, H	entropy (thermodynamic; Shannon)
k_B, T	Boltzmann constant; temperature
$\mathcal{V}$	vacuum manifold (space of energy minima)
π_n	n -th homotopy group
$\mathcal{F}$	a sheaf; $\mathcal{F}(U)$ its sections over the open set U
μ	a measure on the substrate, normalized to total mass one
γ	the rate calibration converting a field estimate to a coupling
τ	time interval over which the contraction rate is evaluated

Part II. Topology, symmetry breaking, and boundaries

3 Spontaneous symmetry breaking

What it is. A physical system is described by an energy functional. A *symmetry* of that functional is a transformation—a rotation, a reflection, a sign flip—that leaves the energy unchanged. Symmetry is *spontaneously broken* when the energy functional has a symmetry that its lowest-energy states do not: the laws are symmetric, the outcomes are not. The standard toy model is the double well

$$V(v) = (v^2 - 1)^2,$$

which is invariant under $v \mapsto -v$ but has two minima, at $v = -1$ and $v = +1$. A ball in this potential must settle in one of them; whichever it picks, the reflection symmetry is gone from the outcome while remaining in the law.

The set of all lowest-energy states is the *vacuum manifold* $\mathcal{V}$. For the double well it is the two-point set $\{-1, +1\}$. In richer models it can be a circle, a sphere, or a more complicated space, and its shape is what governs everything in the next section.

Why the framework needs it. The framework’s first move is that a conscious agent must be a *bounded* thing—something with an inside and an outside, separable from its environment. Rather than assuming a boundary, the framework wants one to be forced by physics. Symmetry breaking supplies the mechanism: if different regions of a substrate settle into different minima, something has to happen in between, and that something is a boundary.

The statement. Two results are proved, and the difference between them matters. The *pointwise vacuum* result says that a field globally minimizing the potential energy lies in the vacuum manifold *almost everywhere*—not everywhere. The almost-everywhere is not slack in the proof; it is necessary. On the double well over a two-point spacetime carrying a Dirac probability measure, the minimizing field sits at a vacuum value on the support of the measure and at 0 off it, so the pointwise version of the claim is false of the very field the theorem applies to.

What it does not give. No conserved quantity is constructed anywhere from this, and there is no Noether theorem in the development. The symmetry group and the metric on the substrate appear in the definitions but are consumed by no theorem.

In Lean. `pointwise_vacuum_of_global_min` and `spontaneous_symmetry_breaking` in `Phase1.Primitives.lean`. The witness in `Examples.lean` §11 instantiates the $\mathbb{Z}_2$ action $v \mapsto \pm v$ on the double well, proves the vacuum manifold degenerate, and proves the two minima exchanged by the symmetry.

Unusually, its minimization hypothesis is *proved* rather than assumed—everywhere else in the development the existence of a global minimizer is a hypothesis, since no compactness or direct-method argument is formalized.

4 Topological defects: the Kibble classification

What it is. Suppose a field takes values in the vacuum manifold $\mathcal{V}$ almost everywhere across some region of space. Different parts of the region may have chosen different vacuum values. Whether the field can interpolate smoothly between them depends on the *topology* of $\mathcal{V}$ —specifically on its homotopy groups (Kibble, 1976):

- $\pi_0(\mathcal{V}) \neq 0$ ($\mathcal{V}$ is disconnected): **domain walls**. Two regions in different connected components cannot be joined without leaving $\mathcal{V}$ somewhere.
- $\pi_1(\mathcal{V}) \neq 0$ ($\mathcal{V}$ has non-contractible loops): **vortices** (line defects).
- $\pi_2(\mathcal{V}) \neq 0$: **monopoles** (point defects).

$\pi_0(\mathcal{V})$ is just the set of connected components—the crudest possible topological invariant, and the only one the framework uses.

Why the framework needs it. A domain wall is a boundary. If a defect is forced, a bounded region is forced, and the framework’s first requirement—a thing separate from its environment—is met by physics rather than by stipulation.

The statement. The result taken is the π_0 case, and it is startlingly cheap:

Let ϕ be a continuous field on a *connected* substrate, taking values in a space M . If $\phi(x_1)$ and $\phi(x_2)$ lie in different connected components of M —equivalently, if no preconnected subset of M contains both—then ϕ takes some value outside M .

The entire proof is: a continuous image of a connected set is connected. The image $\phi(\text{substrate})$ is connected; if it stayed inside M it would be a connected subset of M containing points of two different components, which is impossible. So it leaves M . That exit point is the wall.

Note what is *absent* from the hypotheses: no energy potential, no Lagrangian, no symmetry group, no dynamics. Only continuity and connectedness.

There is a second, differently directed result, the *extension obstruction*: boundary data that is not null-homotopic admits no continuous extension over a contractible interior. This says a defect *obstructs extension*; it does not say a defect must exist. The two directions are easy to conflate and the development keeps them apart deliberately.

What it does not give. Higher defects (π_1 vortices, π_2 monopoles) are recorded as available and not taken; the argument uses domain walls alone. And the theorem forces a *topological* boundary, not an information-processing register—joining those two is exactly the assumption E12 carries, and E12 is classified as an independent physical premise for that reason.

In Lean. `exists_notMem_of_no_common_preconnected` and `exists_notMem_of_connectedComponentIn` (Phase1Primitives.lean §3), built on Mathlib’s `IsPreconnected.subset_connectedComponentIn`. The witness (Examples.lean §11.1) does not merely note that the double well has two minima—it *computes* the vacuum manifold, `wellVacuum.eq` proving `DynamicalVacuum V = {−1, 1}` exactly. That matters: a larger vacuum set might have been connected, and the theorem would not apply. `wellVacuum_separated` then proves the disconnection in the form the theorem consumes, by observing that a preconnected subset of $\mathbb{R}$ spanning -1 and 1 must contain 0 , which is not a minimum. Non-emptiness of the class of fields satisfying the hypotheses is checked rather than assumed: the clipped identity on $\mathbb{R}$ qualifies, and `kink_zero_notMem` locates its forced wall at the origin.

Part III. Information, entropy, and the cost of forgetting

5 Finite capacity

What it is. Shannon entropy $H(p)$ measures, in bits or nats, how much uncertainty a probability distribution carries—equivalently, how much information you gain on average by learning which outcome occurred. On a finite set X it is bounded:

$$H(p) \leq \log |X|,$$

with equality exactly when p is uniform. A system with $|X|$ distinguishable states cannot hold more than $\log |X|$ nats, however cleverly it encodes.

A second, purely combinatorial fact does most of the work. If an environment delivers a stream of k perturbations drawn from an alphabet P , there are $|P|^k$ possible histories. If $|P|^k > |X|$, then by the pigeonhole principle the map from histories to internal states cannot be injective: two distinct histories leave the system in the same state. This holds *whatever* the update rule is, and even when every individual step of it is reversible.

Why the framework needs it. A bounded agent is bombarded by its environment. It cannot retain everything. Something is discarded, and the next section prices that.

What it does not give. Finite capacity is *assumed*, not derived—nothing here proves cortex has a finite effective register. And finite capacity does not imply that every update erases; it implies only that *some* history pair collides once the stream is long enough. The hypothesis is also fenced: on an *infinite* state space the corresponding step fails outright, as `Nat.succ` witnesses—it misses 0 yet is injective.

In Lean. `shannon_entropy.eq_log_card_iff` and `absorb_not_injective` in `Phase1PhaseSpaceCapacity.lean`; the infinite-space refutation is recorded at `Examples.lean` §19.

6 Landauer’s principle

What it is. Landauer (1961) observed that logically irreversible operations have a thermodynamic price. Erasing one bit of information—mapping two distinguishable states onto one—requires dissipating at least $k_B T \ln 2$ joules of heat into the environment. The reasoning: the microscopic laws are reversible, so phase-space volume is conserved (Liouville’s theorem). If the system’s own state space contracts, the volume has to go somewhere, and the only place available is the bath. Bath volume expanding means bath entropy rising, and $\Delta Q = T \Delta S_{\text{bath}}$ converts that to heat.

Why the framework needs it. This is what makes a bounded agent necessarily a *dissipative structure* rather than a passive one, and it supplies the thermal budget that the next part spends.

The statement. Three vocabularies for the same heat appear in the development, and they are deliberately connected rather than left side by side.

Combinatorial.

On a finite state space, a map that fails to reach every state is not injective (`is_eraser_of_not_surjective`) Under globally reversible system–bath dynamics, non-injectivity of the system update forces $\Delta Q > 0$ (`finite_phase_space_dissipates`, `landauer_from_reversibility`). A *second law* statement follows in the same file.

Measure-theoretic.

The same statement with continuous entropy and volume contraction in place of counting arguments. The two are proved to coincide at counting measure on a finite register: `landauer_heat_bound_count_eq`.

Informational.

Wasted memory as erased entropy—see Section 8.1.

The bound is *not* a universal “ $k_B T \ln 2$ per non-injective step”. It is T times the entropy actually erased by the particular update; erasing one unbiased bit is the special case giving $k_B T \ln 2$.

What it does not give. The step from “the state space contracted” to “heat was dissipated” requires the thermodynamic postulate $\Delta Q = T \Delta S_{\text{bath}}$. That is carried as a class field, `StatisticalMechanics.heat_eq`, not proved—and `Examples.lean` exhibits a one-bit erasure system with an explicit bath discharging it, so the postulate is demonstrably satisfiable rather than merely asserted. Identifying such a register with a biological boundary is a further modelling step, and it is exactly what E23 carries.

6.1 The Norton–Shenker distinction

What it is. The standard objection to naive readings of Landauer’s principle (Norton, 2005; Shenker, 2000): *logical* irreversibility—a many-to-one map on abstract state labels—does not by itself entail heat. If the physical implementation is injective, nothing has been erased physically, and no heat need flow. A computation can look many-to-one at the level of the abstraction while the underlying device is perfectly reversible.

Why the framework needs it. Without this distinction the whole thermodynamic argument is a pun on the word “erasure”.

The statement. The development defines erasure as non-injectivity on the *physical* state space, never on abstract labels. The distinction survives explicitly in the model: retaining an incoming perturbation is injective and free (`keepRecord.injective`), while refreshing the register that held it is not (`absorbStep.is_erasure`). In informational form the same distinction reappears as `nonpredictive_eq_zero_of_injective`: on a reversible update, the entropy erased is zero, so the bound forces wasted memory to vanish.

Part IV. Predictive thermodynamics

7 Mutual information and the data processing inequality

What it is. Mutual information $I(X; Y)$ measures how much knowing X tells you about Y . Formally it is the KL divergence of the joint law from the product of its marginals:

$$I(X; Y) = D_{KL}(P_{XY} \parallel P_X \otimes P_Y),$$

which is zero exactly when X and Y are independent.

The *data processing inequality* says information cannot be created by processing. If $X \rightarrow Y \rightarrow Z$ forms a Markov chain— Z depends on X only through Y —then $I(X; Z) \leq I(X; Y)$.

Why the framework needs it. The framework needs to split what a system remembers into a part that predicts the future and a part that does not. The chain here is $X_t \rightarrow S_t \rightarrow S_{t+1}$: the system state X_t at time t was built from the environmental signal S_t , and the environment evolves by its own dynamics, unaffected by the system. Then $I(X_t; S_{t+1}) \leq I(X_t; S_t)$: what the

memory says about the future is at most what it says about the present. The difference,

$$\underbrace{I(X_t; S_t)}_{\text{total memory}} - \underbrace{I(X_t; S_{t+1})}_{\text{predictive memory}} = \text{nonpredictive memory} \geq 0,$$

is well-defined and non-negative.

What it does not give. The Markov structure encodes a *no-back-action* assumption: the system does not act on its environment. This is not decoration. The development fences it with a counterexample—a Markov kernel on the pair that writes the system’s state into the signal and thereby *raises* mutual information, breaking the inequality.

In Lean. `Mathlib` carries no mutual information, so it is defined from `k1Div` in `Phase3.PredictiveThermodynamics`; the inequality is `predictiveInfo.le_mutualInfo`.

8 The Still–Sivak–Bell–Crooks bound

What it is. [Still et al. \(2012\)](#) proved that nonpredictive memory is thermodynamically expensive:

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]$$

The work a system dissipates is at least the thermal cost of the information it holds about the present that is useless for the future. Memory that fails to predict is paid for in heat.

Why the framework needs it. This is the framework’s answer to a real gap. A dissipative structure that merely dissipates is not yet cognitive. The bound supplies a reason for structure to become predictive: under selection pressure to minimize dissipation, useless memory is a cost, so what survives is memory that predicts. The framework calls this *structural resonance*.

The statement. The bound itself is a physical postulate—a class field, `PredictiveDissipation.still_bound`, not a theorem. Everything downstream *is* derived: the data processing inequality above; non-negativity $W_{\text{diss}} \geq 0$ (which follows rather than being assumed); the statement that a given dissipation budget bounds wasted memory; and the statement that a quasi-static drive, dissipating nothing, leaves every retained bit predictive.

A related exact result is also carried: the Kawai–Parrondo–Van den Broeck equality ([Kawai et al., 2007](#)),

$$\langle W_{\text{diss}} \rangle = k_B T D_{KL}(\text{forward} \parallel \text{reverse}),$$

from which `nonpredictive.le_arrow` follows: the irreversibility of the dynamics is at least the memory that fails to earn its keep.

What it does not give. This is the single most over-read result in the framework, and the manuscript is emphatic about it. The bound does *not*:

- specify a learning rule;
- maximize prediction;
- require the internal representation to reproduce environmental geometry;
- entail Bayesian inference or semantic content.

Section 25 reports a simulation making this concrete: a coupling update that removes most of a squared-drift objective maintains coherence while moving *away* from the structure it is expected to learn—further from it than from almost every relabelling of the nodes.

In Lean. `Phase3.PredictiveThermodynamics.lean`. The class is inhabited twice over one correlated two-bit law ([Examples.lean §18](#))—a frozen signal permitted to dissipate nothing, and a scrambled one forced to dissipate its entire memory—so the theorems about it are not vacuous.

8.1 Deriving Still’s bound from Landauer’s

What it is. Two thermodynamic charges have been introduced: Landauer charges a register for *destroying* information; Still charges a structure for *keeping* information it cannot use. They are the same heat viewed twice, and the development connects them.

The statement. `PredictiveDissipation.ofLandauer` builds a predictive structure out of a given register: its thermal scale is that register’s temperature, its dissipated work is that register’s Landauer heat, and Still’s bound is *derived* from `landauer_bound` rather than postulated a second time. The single hypothesis is the identification

$$\underbrace{I_{\text{mem}} - I_{\text{pred}}}_{\text{memory wasted}} \leq \underbrace{S(\text{id}) - S(t)}_{\text{entropy erased}}.$$

What it does not give. That identification is a restriction, not bookkeeping. On an injective update the right-hand side is zero, so the inequality forces the waste to vanish—the Norton–Shenker distinction in informational form. The witness is the one-bit eraser, where a register destroying $\log 2$ pays for a memory wasting $\log 2$, with equality. Connecting the budget to Landauer heat therefore requires nonpredictive information to be bounded by erased entropy; the formal bridge derives the bound *under* that identification and does not derive the identification. That is E34.

9 What does not follow: the free-energy reading

What it is. A tempting inference, and one associated with free-energy formulations (Friston, 2010): if a system minimizes its dissipation, and dissipation bounds the divergence between its internal model and the world, then the divergence must go to zero—the system converges on a true model of its environment.

The statement. The inference is invalid, and the development says exactly where. The assumption available is a bound of the form

$$D_{KL}(P||Q) \leq \Delta t \sigma,$$

carried as the class field `StructuralResonance.kl_bound`. Descending σ does not force $D_{KL} \rightarrow 0$ unless the right-hand side itself tends to zero. A driven system settles into a non-equilibrium steady state with $\sigma > 0$, and the bound then permits *any* divergence up to $\Delta t \sigma$. The witness `boolDetuned` attains the assumed bound with $D_{KL} = \sigma = \log 2$: a positive limiting divergence is permitted, so the limit cannot be read off the bound.

Separately, Gibbs’ inequality proves $D_{KL} \geq 0$. It does not identify relative entropy with physical entropy production. Conflating the two is the other half of the same error.

What it does not give. The framework replaces the invalid inference with the correct, weaker bound and declines the conclusion. It does not claim the free-energy conclusion is false—only that it does not follow from these premises.

Part V. Operators, gradients, and the squared-drift objective

10 The squared-drift functional and its gradient

What it is. Fix the phases and ask how fast the system is being pushed around. Each site x has a drift—the instantaneous velocity its phase is being driven at:

$$v(x) = \omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y).$$

The *squared-drift functional* totals the squares of these:

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x).$$

Read as a function of the kernel K with phases held fixed, this is a *quadratic functional of an affine image of K* : the inner integral is a linear operator applied to K , then ω is added (affine), then the result is squared and integrated (quadratic).

Why the framework needs it. The framework wants a quantity that a plasticity rule could descend, and wants to know whether descending it organizes the coupling usefully.

The statement. Functions of this shape have gradients computable by Hilbert-space methods. Let A be the bounded linear operator sending a kernel to its drift field, and A^* its adjoint—the operator satisfying $\langle Au, w \rangle = \langle u, A^*w \rangle$ for all u, w . Then

$$\nabla \sigma = \frac{2}{D} A^*(AK + \omega),$$

exactly as for a finite-dimensional least-squares problem, and the gradient flow $\dot{K} = -\nabla \sigma$ decreases σ at fixed phases.

A second, more surprising result concerns *symmetric* (reciprocal) kernels. For symmetric K , the integrand $K(x, y) \sin(\theta_y - \theta_x)$ is antisymmetric under swapping x and y , so its double integral vanishes: the *mean* drift is the same for every phase field. Since the mean is pinned, minimizing the sum of squares amounts to minimizing the variance, and a phase-locked field achieves the minimum. Reciprocity is necessary—a one-way two-site coupling strictly beats its own phase-locked state.

What it does not give. Three limitations, all stated in the source. First, σ is *not* in general a thermodynamic entropy-production rate. The Lean identifier is named `entropy_production_rate`, and the supplement notes explicitly that the name does not establish the identification: an actual entropy-production rate requires probability-current information absent from this calculation. Second, descent is proved at *fixed phases*; joint noisy phase-and-coupling dynamics is a different problem, and Section 25 shows the guarantee does not survive it. Third, whether biological couplings follow this flow at all is an additional modelling question.

In Lean. `Phase8.ContinuousField.lean`: `kernelCLM`, `continuumDriftCLM`, the Hilbert-space adjoint, and `structural_resonance_decreases_sigmaContinuum`, with an atomless witness on the unit interval (`Examples.lean` §20). The symmetric-kernel result is `phase_locked_minimizes_entropy_of_symm`, its mean-matching step `total_drift_eq_of_symm`, and its necessity counterexample `Examples.lean` §16.

Part VI. The Kuramoto model: dynamics

11 The model

What it is. The Kuramoto model is the standard minimal model of synchronization. N oscillators each carry a phase θ_i on the circle and an intrinsic frequency ω_i ; each is pulled towards the others by a sinusoidal coupling. The framework uses the *noisy* version:

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here W_i are independent Wiener processes (Brownian motions), so $\sqrt{2D} dW_i$ is white noise of strength set by the *phase diffusion* $D > 0$. The uniform mean-field convention is $K_{ij} = K/N$.

The *order parameter* r_N is the length of the average of the unit vectors $e^{i\theta_i}$. If all phases coincide the vectors add up and $r_N = 1$; if they are spread uniformly they cancel and $r_N \approx 0$. It is the canonical measure of collective coherence, and everything empirical in the framework is expressed through it.

Why the framework needs it. Because it is the simplest dynamics in which a genuine phase transition between incoherence and coherence occurs as a coupling parameter is raised— which is precisely the structure the coupling-gated recovery hypothesis needs.

What it does not give. One modelling point is made repeatedly and is easy to miss. *Frequency heterogeneity and phase diffusion are separate parameters.* Classical Kuramoto analyses often use a spread of natural frequencies ω_i as the disordering influence; the framework's stationary theory assumes *identical* frequencies and gets its disorder from D . A frequency-spread width cannot be substituted for D when applying that theory, and the simulations are run with diffusion rather than a quenched-frequency surrogate for exactly this reason.

12 The Lyapunov potential and gradient descent

What it is. A *Lyapunov function* is a quantity that never increases along a trajectory; finding one is the standard way to show a dynamical system settles rather than wanders. For the Kuramoto system at zero natural frequency the function is

$$V(\theta) = -\frac{1}{2} \sum_{i,j} A_{ij} \cos(\theta_j - \theta_i),$$

and the dynamics is exactly gradient descent on it: the velocity $\dot{\theta}_i$ equals $-\partial V / \partial \theta_i$. Differentiating along a trajectory gives the identity

$$\dot{V} = -\sum_i \dot{\theta}_i^2 \leq 0,$$

so V decreases at a rate equal to the total squared velocity, and stops decreasing only when the motion stops.

Why the framework needs it. This converts a question about trajectories into a question about a landscape, and the landscape can be characterized.

The statement. Two directions are proved, for *positive* couplings: `phase_locked_minimizes_potential` (a phase-locked configuration attains the global minimum) and `potential_min_implies_phase_locked` (any minimizer is phase-locked). Together: minimizers of V are exactly the phase-locked states.

What it does not give. Positivity is not a technical convenience. A network of all-positive couplings is an unfrustrated ferromagnet with one minimum up to a global phase shift—exactly *not* the near-degenerate landscape that a memory argument from geometric frustration would require. The development is careful about where positivity enters:

- Results that assume it: `phase_locked_minimizes_potential`, `potential_min_iff_phase_locked`, the class field `ThermodynamicCover.A_pos` (hence every conclusion at the gluing node), and `kuramoto_tendsto_global_minimum`, which needs the strictly stronger uniform bound $0 < a \leq A_{ij}$.
- Results that do *not*: existence and uniqueness of trajectories, the Lyapunov identity itself, monotone decrease of the potential, its convergence, and $\sum_i \dot{\theta}_i^2 \rightarrow 0$.

So *that* the motion stops holds for frustrated systems too; *where* it stops does not. Because positivity is a field of one structure rather than a hypothesis scattered through the development, it enters the chain at one node and not before it.

13 The rotating frame

What it is. With nonzero natural frequencies the potential V acquires a term $-\sum_i \omega_i \theta_i$, which is linear in the phases and hence *unbounded below*: the potential has no minimum at all, and a minimum-attaining argument is not available. The standard repair is to change to a co-rotating coordinate frame, $\theta_i \mapsto \theta_i - \Omega t$, which subtracts the common rotation.

The statement. For *identical* natural frequencies $\omega \equiv \Omega$, the reduction is exact: a trajectory maps to a trajectory of the zero-frequency system, the two potentials coincide on it, the descent identity transports, and the chain runs through the minimum characterization to phase-locking and $r^2 = 1$. Phase differences, phase-locking and the order parameter magnitude are all proved frame-invariant, so no observable is lost in the change of coordinates.

What it does not give. Exact only for identical frequencies. A spread of frequencies leaves residual detunings $\omega_i - \Omega$ in the reduced system, to which the unboundedness result still applies, and phase-locking then depends on the threshold—which this file does not treat.

In Lean. `Phase4.RotatingFrame.lean`: `kuramoto_potential_unbounded_below`, `is_kuramoto_trajectory_rotate`, `dynamic_potential_descent`, `rotating_frame_chain`. Run end to end on a two-oscillator system in `Examples.lean` §7.

14 Do the trajectories exist? Picard–Lindelöf, globally

What it is. Before asking where a differential equation goes, one must know it has a solution and only one. The Picard–Lindelöf theorem gives both, locally, when the vector field is Lipschitz—meaning it does not change faster than linearly in the state.

The statement. The Kuramoto vector field is *globally* Lipschitz, with constant $2\sum_{ij} |A_{ij}|$: $\sin$ is 1-Lipschitz and the index set is finite, so there is no smallness caveat, and the natural frequencies drop out of the constant entirely. Hence uniqueness on all of $\mathbb{R}$: two solutions agreeing at one instant agree forever, in both time directions.

Existence globally in time takes one extra step, because Mathlib’s `IsPicardLindelof` caps a local solution’s lifetime by the radius of the ball on which the field’s bounds hold divided by the field’s sup-norm there, and carries no global-in-time theorem. The field here is bounded by $\sum_i |\omega_i| + \sum_{ij} |A_{ij}|$ on the *whole* state space, so the cap can be defeated by enlarging the ball *after* choosing the window. Solutions on every window $(t_0 - T, t_0 + T)$ are produced, uniqueness makes the family coherent on overlaps, and they glue into a solution on all of $\mathbb{R}$.

In Lean. `kuramotoField_lipschitz`, `kuramotoField_norm_le`, `kuramoto_exists_on_window`, `is_kuramoto_trajectory` and the combined statement `kuramoto_cauchy_problem`: the trajectory predicate is inhabited exactly once for each initial state.

15 Where the trajectories go: Barbalat and Łojasiewicz

This section describes the longest single chain of analysis in the development, and it is worth following because two of its steps replace textbook tools that Mathlib does not carry.

15.1 The motion stops

What it is. The potential decreases and is bounded below (at zero frequency), so by monotone convergence it converges: $V(\theta(t)) \rightarrow V_\infty$ along *every* trajectory. Integrating the Lyapunov identity gives the quantitative form—the energy dissipated over $[0, T]$ is exactly the potential dropped—so the improper integral

$$\int_0^\infty \sum_i \dot{\theta}_i^2 dt$$

converges, to the total drop.

The statement. Finiteness of that integral is *not* the same as $\sum_i \dot{\theta}_i^2 \rightarrow 0$: an integrable function can spike forever on ever-narrower intervals. The classical bridge is **Barbalat’s lemma** (roughly: a uniformly continuous function with a convergent integral tends to zero). Mathlib does not have it, so it is proved here, in the form: *a non-negative Lipschitz function whose integrals $\int_0^T g$ are bounded above tends to zero*. The proof idea is geometric—a spike of height ε drags a whole window of width $\varepsilon/2K$ up with it, and only finitely many such windows fit under a finite integral.

Applying it needs the dissipation rate to be Lipschitz in time, which is checked without differentiating it: velocities are bounded, the field is Lipschitz in the configuration, and the configuration is Lipschitz in time because its derivative *is* the bounded field. Conclusion (`velocity_sq_tendsto_zero`): along every trajectory of a zero-frequency system, $\sum_i \dot{\theta}_i^2 \rightarrow 0$.

What it does not give. This locates no limit. It says the motion stops, not where.

15.2 Where it stops

What it is. Locating the limit looks like a job for the *LaSalle invariance principle*, which Mathlib does not carry, and which would want a compact invariant set that the state space $V \rightarrow \mathbb{R}$ does not supply. Neither turns out to be needed. LaSalle is what one needs when a decreasing functional is *all* one has; here the flow is a gradient flow whose minimizers are already characterized (Section 12), and the right tool is a *Łojasiewicz* (or Polyak–Łojasiewicz) estimate: an inequality bounding how far the functional is above its minimum by how fast it is currently decreasing.

The statement. The estimate proved is: on the region where every pairwise phase difference lies within a quarter turn,

$$2a^2W \leq \left(\sum_{ij} A_{ij} \right) \sum_i \dot{\theta}_i^2,$$

where W is the excess of the potential over its minimum and $a > 0$ is a lower bound on the couplings. Its proof uses no spectral gap and no convexity—only two extremal indices, the sites carrying the largest and smallest phase.

Combined with $\dot{W} = -\sum_i \dot{\theta}_i^2$ this is the differential inequality $\dot{W} \leq -cW$ with $c = 2a^2 / \sum_{ij} A_{ij}$, which integrates elementarily (“ $W(t)e^{ct}$ has non-positive derivative”) to exponential decay $W(t) \leq W(0)e^{-ct}$. The rate transfers to the velocity via $|\sin d| \leq \sqrt{2(1 - \cos d)}$,

giving $|\dot{\theta}_i(t)| \leq Be^{-ct/2}$, which is integrable on $(0, \infty)$; since $\theta_i(t) = \theta_i(0) + \int_0^t \dot{\theta}_i$, the trajectory itself converges, and $W \rightarrow 0$ forces the limit to be phase-locked—a *global* minimizer with $r^2 = 1$.

15.3 The compactness objection, met without an invariant region

The quarter-turn region has to be shown forward-invariant, or the estimate applies only at the initial instant. It is, and the argument is neat: because all terms of the excess are non-negative, a single pair satisfies $\frac{1}{2}a(1 - \cos(\theta_j - \theta_i)) \leq W$, so an initial excess below $a/2$ —which monotone decrease preserves—keeps every pairwise cosine strictly positive for all time. A phase difference escaping $[-\pi/2, \pi/2]$ would have to cross $\pi/2$ in absolute value by the intermediate value theorem, and there the cosine vanishes. This is branch-pinning by continuity on a connected half-line, and it replaces a quotient by the phase-shift symmetry.

What it does not give. The two hypotheses are on the *initial data* and neither can be removed: splay and twisted configurations are equilibria too, so “every trajectory reaches the phase-locked state” is simply false. What is proved is the arc-condition version. It is witnessed on three sites (Examples.lean §17), where no closed-form solution exists—the trajectory is produced by the existence theorem and its limit by the convergence theorem, the whole numerical content of checking the hypotheses being $\cos \frac{1}{2} > \frac{3}{4}$.

16 The open gap: propagation of chaos

What it is. Everything in Part VII is about a *mean-field* description—a single density over phases, as though each oscillator felt an averaged influence from an independent population. Justifying that from the finite- N system (1) is the problem of *propagation of chaos* (Sznitman, 1991): showing that if oscillators start independent they stay asymptotically independent as $N \rightarrow \infty$, so the joint law factorizes and the one-particle marginal obeys a closed equation.

Why the framework needs it. Without it, the stationary theory of Part VII and the finite system of Part VI are two different objects with a suggestive resemblance.

The statement. No theorem in the development closes this gap, and the manuscript says so plainly. Exchange symmetry alone does not justify factorization, particularly for an ensemble with an unpinned collective phase.

What it does not give. Two finite- N controls probe the gap without closing it, and both are reported as negative or partial:

- **Subcritical.** As N grows, mean order falls toward the finite-size floor and absolute circular pair correlation decreases—consistent with independence. But the matched-sample sliced-Wasserstein distance between the joint law and the product of its marginals barely moves, exposing that statistic’s finite-sample floor rather than measuring convergence.
- **Supercritical.** Mean order stabilizes, but lab-frame pair correlation stays substantial rather than following the $1/N$ reference. The reason is structural, not a failure of the simulation: above threshold the ensemble has an *unpinned collective orientation*, a shared latent variable that every realization draws once. Unconditional independence and independence *conditional on that orientation* are different claims, and only the first is refuted. In the co-rotating frame—which removes the shared orientation—the $N = 1000$ density matches the von Mises target closely.

This is a negative result for the *unconditional* protocol and settles nothing about the conditional mean-field limit.

Part VII. The synchronization transition

17 The von Mises stationary density

What it is. The *von Mises* distribution is the circular analogue of the Gaussian:

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}.$$

It is peaked at ψ with *concentration* $a \geq 0$; at $a = 0$ it is uniform, and as a grows it tightens. The normalizing constant involves I_0 , a *modified Bessel function of the first kind*—the functions that arise when one integrates $e^{a \cos \theta}$ against powers of $\cos \theta$.

Why the framework needs it. It is the stationary density of the noisy mean-field Kuramoto dynamics: the shape the phase distribution settles into when a coherent mean field competes with diffusion. Every distributional prediction the framework makes flows from it.

The statement. For identical frequencies in the rotating frame, the assumed stationary density is the one above with

$$a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}. \quad (2)$$

The second equation is a *self-consistency* condition and deserves unpacking. The concentration a is proportional to the coherence r , because a more coherent population exerts a stronger mean field. But the coherence is itself determined by the density: the average of $e^{i\theta}$ under a von Mises law at concentration a is exactly $R(a)$. So r must reproduce itself, and the fixed points of $r \mapsto R(Kr/D)$ are the possible stationary states. [Sakaguchi \(1988\)](#) is the source for the threshold this yields.

What it does not give. The density is assumed, not derived. Going from the stochastic dynamics to this density would require the spectral theory of the associated Fokker–Planck operator, which Mathlib does not carry. The algebraic step from the stationary Fokker–Planck equation to the von Mises form is verified numerically in the repository (`simulations/fokker_planck_von_mises.py`), not as a theorem. This is the largest single formalization gap in the stationary theory, and it compounds with the propagation-of-chaos gap of Section 16: one connects the finite system to a mean-field density, the other identifies *which* density.

In Lean. `Phase8.SelfConsistency.lean`. Because Mathlib has no modified Bessel functions, `besselRatio` is written as a ratio of interval integrals. `vonMisesDensity` is proved a probability density on $[-\pi, \pi]$; `circularOrderParameter_vonMises` proves its order parameter equals $R(a)$ (real, because $\mathbb{E}_a[\sin \theta] = 0$ by symmetry); and `fixedPoint_iff_selfReproducing` states the equivalence— r solves $r = R(Kr/D)$ exactly when the density it induces has order parameter r .

18 Why the threshold is $K_c = 2D$

What it is. The claim is that coherence is impossible at or below $K = 2D$ and, above it, takes exactly one positive value. The number 2 is not fitted; it falls out of a single inequality about the von Mises family.

The statement. The key algebraic identity is integration by parts:

$$\int_{-\pi}^{\pi} \cos \theta e^{a \cos \theta} d\theta = a \int_{-\pi}^{\pi} \sin^2 \theta e^{a \cos \theta} d\theta,$$

exact because $\sin(\pm\pi) = 0$ kills the boundary term. Dividing by the normalizer, this factorizes

the Bessel ratio:

$$R(a) = a E(a), \quad E(a) := \mathbb{E}_a[\sin^2 \theta].$$

Now everything turns on E . Since $\sin^2 \leq 1$ trivially, $R(a) \leq a$, which would put a threshold at $K = D$. The sharper fact is

$$E(a) \leq \frac{1}{2}, \quad \text{strictly, for } a > 0,$$

which gives $R(a) < a/2$ and hence a threshold at $K = 2D$. **That factor of 1/2 is exactly what separates a threshold at D from one at $2D$,** and the development keeps the crude version around to make the point visible.

Proving $E(a) \leq 1/2$ amounts to showing the second cosine moment $I_2(a) \geq 0$, done by *folding* the interval $[-\pi, \pi]$ onto $[0, \pi/4]$ through the two reflections $\theta \mapsto \pi - \theta$ and $\theta \mapsto \pi/2 - \theta$, after which the integrand becomes a product of two manifestly non-negative factors. Strictness for $a > 0$ comes from positivity of the folded integrand on the interior of that interval.

With $R(a) = aE(a)$ in hand, the two halves of the threshold follow:

Subcritical ($K \leq 2D$).

The only $r \geq 0$ solving $r = R(Kr/D)$ is $r = 0$. Because $R(a) < a/2$, the map $r \mapsto R(Kr/D)$ lies strictly below $r \mapsto (K/2D)r \leq r$ for $r > 0$, so it cannot cross. The threshold case $K = 2D$ falls on the *incoherent* side, so the two regimes exhaust $K \geq 0$ with no gap.

Supercritical ($K > 2D$).

There is an r with $0 < r \leq 1$ solving it, and no series expansion is needed. Writing the condition through E : since $D/K < 1/2 = E(0)$ and E is continuous, the map overshoots for small $r > 0$; $R(a) \leq 1$ (being a mean of $\cos \theta$) makes it undershoot at $r = 1$; and the intermediate value theorem supplies the crossing.

Three further results complete the picture:

- **Uniqueness.** At most one positive solution exists—see Section 19.
- **Continuity at threshold.** For every $\varepsilon > 0$ there is $\delta > 0$ such that on $(2D, 2D + \delta)$ every coherent solution has $r < \varepsilon$. The branch emerges *continuously* from zero: no discontinuous jump. (The proof needs no monotonicity of E —on a compact interval bounded away from the origin the extreme value theorem gives a maximum strictly below 1/2, and beyond that the crude bound $E(a) \leq 1/a$ suffices.)
- **Strict growth.** r increases strictly with K above threshold.

Together: the non-negative solution set is exactly $\{0\}$ at or below threshold and $\{0, r\}$ above it. This is a *pitchfork bifurcation*, and zero remains a solution above threshold—it merely ceases to be the only one.

What it does not give. Everything here is a statement about a stationary *density*. Of the coherent branch the development proves existence, uniqueness, continuous emergence and strict growth—but **not that it is dynamically selected**. No theorem in these files mentions a trajectory, so none could. The order parameter reached is that of a density, not of a trajectory, and no theorem relates the two—that relation is the mean-field limit of Section 16.

There is also a modelling caution about the spatial version. Replacing the finite sum by an integral against a kernel $K(x, y)$ requires the substrate measure to be *normalized to total mass one*, and this is load-bearing rather than cosmetic. The predicate compares the kernel's double average against $2D$; on a substrate of total mass m that average carries a factor m^2 , so without normalization the comparison tracks the substrate's *size* rather than its coupling and can be satisfied by any kernel at all. `Examples.lean` §9 exhibits a system a factor of two below threshold whose unnormalized double integral nonetheless clears $2D$. Reduction of a

heterogeneous spatial kernel to a scalar K is a further modelling step, and the threshold does not automatically transfer to every spatial or signed network.

19 The fold-and-cross proof of uniqueness

This is the one proof worth describing in outline, because its structure is unusual and the supplement devotes a section to it.

What it is. Uniqueness of the coherent branch reduces to the *injectivity* of E on $(0, \infty)$, and what is proved is the stronger statement that E is *strictly decreasing* on $[0, \infty)$.

The statement. The obvious route is to differentiate. For an exponential family, tilting in the natural parameter gives

$$E'(a) = -\text{Cov}_a(\cos^2 \theta, \cos \theta),$$

and one would show that covariance positive. That route was not taken, and it was the harder one: it costs differentiation under the integral sign, and the covariance is not sign-definite pointwise—symmetrizing with an independent copy gives $\frac{1}{2}\mathbb{E}[(X - X')^2(X + X')]$ with $X = \cos \theta$, an integrand changing sign with $X + X'$.

Instead the target is kept as the two-point inequality $C_2(a)Z(b) < C_2(b)Z(a)$ for $a < b$, and two moves settle it.

Fold first. The same two reflections used for $I_2 \geq 0$ carry both integrals onto $[0, \pi/2]$, replacing $e^{a \cos \theta}$ by $W(a, \theta) = 2 \cosh(a \cos \theta)$ and confining $\cos \theta$ to $[0, 1]$. There, $\cos 2x - \cos 2y$ and $W(b, x)W(a, y) - W(a, x)W(b, y)$ have the *same sign*, both being governed by $\cos x$ against $\cos y$. This is the same $X \mapsto -X$ symmetrization the covariance form needed, performed once at the start rather than inside a double integral. Its entire analytic content is one identity: $\cosh(ap)\cosh(bq) \leq \cosh(bp)\cosh(aq)$ for $0 \leq a \leq b$, $0 \leq q \leq p$, which follows from $2 \cosh X \cosh Y = \cosh(X+Y) + \cosh(X-Y)$ and $\cosh$ increasing in $|\cdot|$.

Then cross. Rather than integrating that inequality over a square—which would need a product measure and Fubini—let α be the mean of $\cos 2x$ under $W(a, \cdot)$, and let $x_0 \in [0, \pi/2]$ be a point where $\cos 2x_0 = \alpha$, supplied by the intermediate value theorem. Evaluating at $y = x_0$ and integrating in x alone gives the conclusion, because the term carrying $W(b, x_0)$ vanishes by the choice of α . Strictness comes from positivity of the integrand on a subinterval missing x_0 .

Nothing in the proof differentiates anything. No product measure and no Fubini appear.

The same crossing argument run with $\cos \theta$ in place of $\sin^2 \theta$ needs no fold at all— $R(a) = \mathbb{E}_a[\cos \theta]$ is the mean of the very variable the family is tilted by—and yields `besselRatio_strictMono`: the Bessel ratio R is strictly increasing on all of $\mathbb{R}$. Mathlib carries this in no form.

The statement. The consequences are the uniqueness results (`coherent_fixed_point_unique`, `supercritical_fixed_point_existsUnique`, `supercritical_solution_set`) and strict growth of the branch (`coherent_branch_strictMono`), which uses both halves: E decreasing puts the concentrations of two couplings in the order $a_1 < a_2$ through $E(a_i) = D/K_i$, and R increasing carries that order to $r_1 < r_2$.

What it does not give. Because nothing is differentiated, E and R are obtained as strictly monotone rather than as smooth. The rate at which the branch leaves zero therefore needs a different file—the next section.

Part VIII. The critical exponent and the square-root onset

20 The exponent $\beta = 1/2$

What it is. Near a continuous phase transition, the order parameter typically behaves like a power of the distance to threshold: $r \sim (K - K_c)^\beta$. The exponent β is a *universal* feature of the transition—it does not depend on microscopic details, only on the class the transition belongs to. For mean-field theories $\beta = 1/2$, giving a square-root shape.

Why the framework needs it. This is the source of the framework’s most distinctive falsifiable prediction. A square root has *infinite initial slope*: at the first instant it rises faster than any exponential, logistic or linear function. If recovery from awakening is gated by a threshold crossing, its onset should show that shape—and that shape is not what generic saturating nonlinearities produce.

The statement. Getting the rate costs the differentiation the uniqueness proof avoided.

1. **Differentiate under the integral.** Define the cosine moments $\text{vmMoment } n a = \int_{-\pi}^{\pi} \cos^n \theta e^{a \cos \theta}$. Differentiating in a raises the cosine power by one. The interchange of derivative and integral is justified by dominated convergence, dominated by the constant $e^{|a|+1}$ on a unit ball around the parameter—this is where compactness of $[-\pi, \pi]$ and boundedness of $\cos$ do the work. Iterating gives two continuous derivatives.
2. **Rewrite E through the moments.** $E(a) = 1 - \text{vmMoment } 2 a / \text{vmMoment } 0 a$, a quotient with positive denominator, so E inherits two derivatives.
3. **Evaluate at zero.** The moments at $a = 0$ are elementary integrals of powers of $\cos$: $2\pi, 0, \pi, 0, 3\pi/4$. They give

$$E(0) = \frac{1}{2}, \quad E'(0) = 0, \quad E''(0) = -\frac{1}{8},$$

the first derivative vanishing by the symmetry that kills the odd moments.

4. **Taylor.** Hence

$$E(a) = \frac{1}{2} - \frac{a^2}{16} + o(a^2).$$

The remainder is $o(a^2)$, not $O(a^4)$; the stronger form would need a fourth derivative and nothing requires it.

5. **Substitute into the branch equation.** On the coherent branch $E(Kr/D) = D/K$. Writing $a = Kr/D$ and rearranging gives

$$K - 2D = Ka^2 \left(\frac{1}{8} - 2q(a) \right)$$

with q the normalized remainder. Letting $K \rightarrow 2D$ along *any* family of positive coherent solutions with $r \rightarrow 0$ sends $q \rightarrow 0$ and yields

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad \text{i.e.} \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

The statement is deliberately *filter-indexed* rather than parameterized by a curve, so it constrains every such family and privileges no path. Because its hypotheses hold only *eventually*

along the filter, and an eventual clause is vacuously true on the empty filter, the file also exhibits a family satisfying it: `coherentBranch` selects the unique positive solution above threshold, `coherentBranch_tendsto_zero` reads the continuity result as a limit along $K \downarrow 2D$, and `coherentBranch_critical_exponent` discharges every hypothesis on it. Without that witness the theorem could be true for want of any instance.

What it does not give. Asymptotics of the solutions of a *stationary* equation. It establishes no stability, no dynamical selection and no time course. Reading it as a square-root foot on a recovery curve requires two further assumptions, both stated in the manuscript as assumptions: adiabatic separation, and a threshold crossing at non-zero speed.

In Lean. `Phase8CriticalExponent.lean`: `hasDerivAt_vmMoment`, `contDiff_two_vmMoment`, `vonMisesSRatio_second`, `coherent_solution_critical_exponent`.

21 From exponent to prediction

The statement. If the coupling $K(t)$ rises through K_c at time t_0 with non-zero speed $\dot{K}(t_0) > 0$, and the phase distribution *tracks* the coherent stationary branch, then substituting $K(t) - K_c \approx \dot{K}(t_0)(t - t_0)$ into (3) gives

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0. \quad (4)$$

The ideal picture is a flat subthreshold interval followed by a continuous square-root onset.

What it does not give. Four qualifications, each of which the manuscript raises before a referee could.

1. **No saturation follows.** Where the curve goes after onset requires a specified subsequent coupling trajectory. Monotone coupling alone does not determine a logistic or any unique sigmoid.
2. **Latency is not parameter-free.** The onset time is fixed by $K(t_0) = K_c$ only if $K(t)$ is *independently* known. When coupling is inferred from the same coherence trace, the prediction of latency is circular.
3. **Critical slowing down.** Near a continuous transition, relaxation times diverge, so the adiabatic (branch-tracking) assumption is worst exactly where the prediction is sharpest.
4. **Finite-size floor.** A finite population carries non-zero fluctuating order below threshold, of order $1/\sqrt{N}$. An observable onset therefore requires a *predeclared* threshold above that floor, and a finite-rate trajectory need not show an infinite-slope cusp at all.

Part IX. Finite populations, finite ramps, and what simulation adds

The stationary theory describes an infinite population tracking a branch adiabatically. Cortex is neither infinite nor slow. The simulations exist to map the gap, and their headline results *restrict* the hypothesis rather than support it.

22 Euler–Maruyama integration

What it is. The numerical scheme for stochastic differential equations. The deterministic Euler step $\theta \mapsto \theta + f(\theta) \Delta t$ acquires a noise term scaling as $\sqrt{\Delta t}$ rather than Δt —because Brownian increments over a step of length Δt have standard deviation $\sqrt{\Delta t}$. Applied to (1):

$$\theta_i \mapsto \theta_i + \left[\omega_i + \sum_j K_{ij} \sin(\theta_j - \theta_i) \right] \Delta t + \sqrt{2D \Delta t} \xi_i, \quad \xi_i \sim \mathcal{N}(0, 1).$$

What it does not give. Every simulation reported is a finite run at sampled parameters, with explicit seeds and saved compact summaries. Equal-duration half-step controls probe numerical sensitivity near transitions; they do not establish convergence of the integrator, and no run discharges a hypothesis of the formal composition.

23 Ramp speed sets the apparent onset exponent

What it is. Drive the coupling upward at constant speed $v = \dot{K}$ through the threshold and watch when coherence escapes the floor. The stationary theory says the order parameter should follow the branch; a finite ramp cannot, because the system needs time to relax and the relaxation time diverges at the transition. The result is a *delay*: escape happens at a coupling above K_c , by an amount that grows with v . This is the same physics as the Kibble–Zurek mechanism for driven transitions.

The statement. With $N = 2000$, $D = 1$, 32 seeded replicas and $v \in \{10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}\}$, escape defined as the first of three consecutive samples at $r \geq 0.2$ after threshold:

- The fastest leg is right-censored—only 5/32 replicas escape before $K = 2.5$.
- The three slower ramps give mean coupling delays 0.1947, 0.0684 and 0.0253, whose log–log fit is $\Delta K \propto v^{0.443}$ —near the predicted exponent 1/2 at this resolution.
- At fixed $v = 10^{-2}$ the delay grows with population size (0.0619, 0.1947, 0.2341 at $N = 500, 2000, 8000$), qualitatively matching a predicted $\sqrt{\log N}$ dependence. Three sizes are evidence for a trend, not a scaling law.
- **The apparent onset exponent**, fitted as $r = A(K - K_0)^{\beta_{\text{eff}}}$ over $0.1 \leq r \leq 0.4$, runs 0.964, 0.898, 0.471, 0.417 from fastest to slowest ramp. Slow ramps recover the square-root foot; fast driving pushes the observable foot toward a *linear* apparent onset.

Why this matters. This is a methodological result with reach beyond the framework. **The exponent recovered by fitting a driven transition is a joint function of the mechanism and the drive rate.** An onset exponent read off a recovery trace therefore says little about the underlying transition unless the ramp rate is independently known and the finite-size floor is declared—and that holds for *any* analysis reading a critical exponent off a driven transition.

The first thing it constrains is the framework’s own prediction: the stationary square-root law cannot be imposed on an arbitrary recovery trace, and its absence is not by itself evidence against a coupling-gated mechanism. A linear apparent onset counts against the adiabatic branch interpretation only in a regime where that approximation is independently justified.

What it does not give. The (a, r) collapse is more robust than the temporal shape: independently estimated (a, r) traces stay close to $I_1(a)/I_0(a)$ even where the trajectory lags the branch, with post-critical RMS deviations 0.00245, 0.00599, 0.00689, 0.00705, none reaching twice the slowest run’s floor ($2\text{Dev}_0 = 0.01410$). The specified deviation-based critical ramp speed is therefore *not bracketed* by this sweep. Mapping these dimensionless rates to human recovery times needs an independently calibrated phase diffusion and coupling trajectory.

24 Spatial coupling and geometric frustration

What it is. Two things could break coherence in a spatially extended sheet: coupling that decays too fast with distance to hold the sheet together, and *frustration*—signed (excitatory and inhibitory) couplings that cannot all be satisfied at once, in the way that three mutually antagonistic spins on a triangle cannot.

The statement. *Spatial decay.* On a periodic 128×128 sheet spanning 2.0 mm, an exponential kernel row-normalized to fixed total coupling at every decay length: with steady $r > 0.2$ as the operational criterion, the smallest interpolated decay length above which every larger sampled length stayed coherent was 0.0140 mm. At the empirical controls 0.1, 0.2 and 0.3 mm, steady order was 0.9424, 0.9427 and 0.9427, with winding-defect density at most 2.4×10^{-6} . So this sweep shows no fragmentation in the empirical band—which sits inside a plateau where steady order spans only 0.00038 out to the longest range sampled. The boundary itself is 0.90 lattice spacings, below that grid’s own resolution; halving the spacing—a 256×256 sheet over the same 2.0 mm—leaves it at 0.01413 mm, 1.01 times its coarse value and 1.81 spacings on the finer sheet. So the boundary is a physical length, not a lattice artifact.

Frustration. A row-balanced Dale-law network (no self-edges, Erdős–Rényi edge probability 0.2, 80/20 excitatory/inhibitory presynaptic fractions, each row’s positive and negative weights summing to $\pm g$) shows that row balance *alone* does not guarantee a floor-level incoherent baseline: signed coupling can amplify finite-size fluctuations, and $N\langle r^2 \rangle$ stays above the independent-phase value of unity. In a separately specified weak-synapse regime, adding a uniform per-pair modulatory term ϵ (contributing mean-field strength $K_{\text{eff}} = N\epsilon$) produces an operational coherence crossing at $K_{\text{eff}}/D \in [1.96, 1.99]$ across seeds. The ϵ grid is geometric with ratio 1.136, refined linearly to steps of 0.025 inside the step those values interpolate across; the refined bracket $[1.958, 2.009]$ still contains $K_c/D = 2$.

What it does not give. Each is one finite parameter regime. Fixing total coupling while changing range tests *fragmentation*, not whether cortex supplies that total coupling. The crossing range describes finite-run variability, not a confidence interval or an infinite-size critical point. And the ratio K_{eff}/D is dimensionless—carrying it over to cortex needs the rate calibration γ of Part XVI, which neither sweep supplies. No clinical mechanism is inferred from any of this.

25 Squared-drift plasticity does not learn structure

What it is. It is tempting to read the predictive-dissipation bound as saying that a system lowering its dissipation thereby comes to mirror the structure of its input. The bound licenses no such reading: it bounds *non-predictive* memory, and a memory carrying no information at all satisfies it with room to spare, so lowering dissipation presses toward *less* representation rather than toward better-aligned representation. This control exhibits a system in which the pressure is not merely toward less but away.

The statement. Phases evolve with noise and clustered frequencies (three nearly equal clusters at 0.5, 1, 1.5 per unit time, with non-zero grand-mean drive), while symmetric couplings update every 50 fast steps by descending the squared-drift objective of Section 10. Couplings stay non-negative, hollow, and normalized to fixed total resource. The symmetric-edge derivative of $\frac{1}{2} \sum_i v_i^2$ is $(v_i - v_j) \sin(\theta_j - \theta_i)$, so both endpoints of an edge contribute. Each seed runs three arms on one phase-noise stream: the gradient step, a symmetric random step of the same Frobenius norm, and a frozen kernel.

Denying an implication means exhibiting its antecedent, so the first thing to measure is how much of the objective the update actually removes. Symmetric coupling cancels from the mean

drift, which fixes $\sum_i v_i$ and so puts a floor of $N\langle\omega\rangle^2/D = 990$ under the objective: the update can equalize drift across nodes but never remove it. Across three seeds:

- The frozen arm sits at 1380.6063 to 1382.3448; the gradient arm reaches 1110.5659 to 1116.6277, removing 0.6773 to 0.6914 of the headroom between the two. **So the objective really is being minimized.** The matched random arm removes -0.0173 to -0.0103 , so this belongs to the gradient direction and not to displacing a kernel at fixed resource.
- Second-half mean order lies between 0.9167 and 0.9222. **So coherence survives the descent.**
- The ratio of mean within-cluster to mean between-cluster coupling—a measure unchanged by rescaling the kernel—falls from 0.9824–0.9948 to 0.3507–0.6135, while the random arm leaves it at 0.9965–0.9997. Against 2000 node relabellings the final kernel is further from the true template than from a fraction 0.9510 to 1.0000 of them.

The objective descends by most of what is available to it, coherence is maintained, and within-cluster coupling falls to a fraction of between-cluster coupling. **Coherence is not a proxy for representational alignment in this regime, and descending a squared-drift functional does not produce representational learning.**

Nor is the direction of the failure accidental, and this is the part worth carrying away. Because symmetric coupling conserves the mean drift, minimizing $\sum_i v_i^2$ can only *equalize* drift across nodes—and equalizing drift rewards coupling between clusters of *unlike* frequency, which is the complement of a template rewarding coupling within clusters of *like* frequency. Objective and template are opposed by construction. That is not a weakness of the demonstration but its content: nothing about a system’s dissipation fixes which structure its slow variables acquire, and which functional they follow is what decides it.

What it does not give. The template is a covariance of shared latent cluster signals, used as a diagnostic—it is not an input to the update, nor a measured covariance of the constant frequencies. Two readouts are recorded but do not carry the result. Frobenius distance to the template grows, but the kernel norm grows by a factor 12.4094 to 14.6958 over the same interval and a fixed *total* does not hold the *norm* fixed, so most of that growth is mass concentrating onto fewer edges rather than misalignment; this is why the scale-free ratio is the statistic quoted above. The off-diagonal correlation with the template ends at -0.0362 to -0.0208 , near where it started: at this sparsity edge-level variance swamps a block-mean contrast. The learning rate 0.2 is chosen by a sweep from 0.001 to 0.4 as the rate at which the objective descends furthest—a declared selection, not a rate at which the update is known to converge. Resource constraints and evolving phases remove the fixed-phase monotonicity guarantee of Section 10, so this is not a counterexample to that theorem but a demonstration of its scope. A plateau in the objective establishes neither stationarity of the adaptive process nor a thermodynamic entropy-production rate—the latter would require probability-current information the calculation does not contain. And the non-zero grand-mean frequency drives the collective coordinate, since symmetric interactions cancel in its mean drift.

Part X. From discrete to continuum

26 Building a coupling matrix from a field

What it is. The framework wants the coupling matrix A_{ij} to be *derived* from a continuum physical field rather than posited. The construction: triangulate the substrate; assign to each edge (u, v) a region of the manifold; and define the coupling weight as the integral, over that region, of a scalar magnitude read off the stress–energy tensor. Symmetry and non-negativity of A are then theorems rather than assumptions.

The statement. The bridge back to the continuum is *mesh refinement*: as the regions shrink, the discrete energy $\frac{1}{2} \sum_{u,v} \mu(R_{uv}) f(x_u)$ should converge to $\int_S f$.

Stated over the bare class of triangulations, that is **false**—nothing forces the regions to cover the manifold, empty regions have diameter zero, and the everywhere-empty triangulation refutes it. Stated correctly it is a theorem, quantified over sequences of triangulations whose regions are measurable, pairwise disjoint across distinct edges, covering, and anchored to the embedded vertices, with mesh size tending to zero. The fixed-mesh error is bounded by $\varepsilon \mu(S)$ with ε a modulus of continuity of the integrand, and the symmetric double sum is proved equal to the midpoint rule over unordered edges.

Carrying the regularity conditions as a structural field, rather than checking them case by case, buys three further statements:

- **weight_self**: a vertex has no edge to itself, so the coupling matrix has zero diagonal.
- **face_of_weight_ne_zero**: a non-zero weight forces $\{u, v\}$ to be an actual face of the complex—the discretization cannot invent couplings between vertices the triangulation does not join.
- **total_weight_eq_setIntegral**: $\frac{1}{2} \sum_u \sum_v w(u, v) = \int_S |T|$ —total coupling strength is exactly the scalar stress-energy of the region covered. Nothing counted twice (disjointness), nothing dropped (covering).

That last identity is the precise sense in which the coupling matrix is derived from the continuum rather than posited.

What it does not give. Three limits. (i) The theorem gives *convergence*, not the $O(1/N^2)$ rate the numerics report. (ii) One field is deliberately left free and marked as a modelling choice: **scalar_magnitude**, the scalar read off the stress-energy tensor, constrained only to be non-negative. A $(0, 2)$ tensor has no canonical scalar—the trace, an observer’s energy density and the tensor norm are different functions, and the structure carries neither a metric nor a frame—so imposing one would pin a choice rather than derive it. That this is a genuine physical input is stated as a *theorem*: two instances over the same triangulation, tensor and regions, differing only in that field, satisfy the partition identity while every edge weight differs (**gridThermo_magnitude_matters**). (iii) The witness is one-dimensional with a constant stress-energy tensor, so it establishes satisfiability, not a physically interesting discretization.

27 The negative result: coarse-graining produces no kernel

What it is. It would be natural to read the mesh-refinement theorem as constructing the continuum kernel $K(x, y)$ that the rest of the framework uses. It does not, and the development proves this rather than noting it.

The statement. Three obstructions, all recorded as results:

1. **mesh_refinement_convergence** converges a *scalar* coupling energy. It produces no kernel and no field dynamics.
2. The discrete side has no product structure to pass to a limit: **edge_region** is a subset of M , not of $M \times M$, so there is nothing kernel-shaped there to converge.
3. The obvious repair—a kernel supported on the vertices—does carry the weights, and yet contributes *exactly zero* to the continuum coupling energy on an atomless substrate.

Point (3) needs a word on *atomless* measures. A measure is atomless when no single point carries positive mass—Lebesgue measure on an interval is the example. A kernel supported on finitely many points is then integrated over a set of measure zero in each argument, so its double integral against $\mu \otimes \mu$ vanishes identically, for every phase field and every choice of weights.

What it does not give. The manuscript is scrupulous about the direction this cuts. It is *Derivation 8’s hardware no-go* (Part XIII) applied to the framework’s own coarse-graining

step—the same argument that would separate a finitely-sited architecture from a continuum one also blocks the framework’s own route from mesh to kernel. The continuum kernel is therefore a *modelling input*, and its identification with the brain’s endogenous electromagnetic field is a further empirical commitment. E45 (dissipation $\rightarrow$ continuum) and E56 (continuum $\rightarrow$ field) are where those two steps live.

Part XI. Sheaves, gluing, and the unity of a state

28 What a sheaf is

What it is. A *presheaf* assigns data to each open region of a space, together with *restriction maps* that cut data down to smaller regions. If $\mathcal{F}(U)$ is the set of “local descriptions over U ” and $V \subseteq U$, there is a map $\mathcal{F}(U) \rightarrow \mathcal{F}(V)$ discarding what happens outside V . In this framework the data are probability measures: each cortical patch carries a distribution over what it represents.

A presheaf is a *sheaf* when local data that agree on overlaps assemble uniquely into global data. Formally: given an open cover $\{U_i\}$ of X and sections $P_i \in \mathcal{F}(U_i)$ such that P_i and P_j have the *same restriction* to every intersection $U_i \cap U_j$, there is exactly one global section $P_X \in \mathcal{F}(X)$ restricting to each P_i . This is the *gluing axiom*, and it is the whole reason sheaves appear here.

The auxiliary notion is the *stalk* at a point x —the space of *germs*, obtained as a colimit over all neighbourhoods of x , capturing what the data look like arbitrarily close to x .

Why the framework needs it. Because synchronization alone does not give unity, and the framework needs a mechanism that assembles local states into *one* state. Sheaf gluing is that mechanism, and it delivers uniqueness for free.

The statement.

Theorem (Sheaf-theoretic unity). Given a sheaf $\mathcal{F}$ of probability measures over a substrate X , if local states $P_U \in \mathcal{F}(U)$ agree on all intersections $U \cap V$, they glue uniquely into a global section $P_X \in \mathcal{F}(X)$.

Both existence and uniqueness are derived from the local data. The construction uses local data only—no global state is supplied as a premise—and the statement that every local section is the restriction of the glued one is a *theorem* about the object produced, not a hypothesis.

29 Compatibility is not phase agreement

This is the conceptual hinge of the whole framework, so it is worth stating slowly.

What it is. Take two cortical regions whose receptive territories overlap. Each carries a probability distribution over what occupies the shared part. Their oscillators can be locked to the *same phase* at every instant while one region’s distribution places an object left of fixation and the other places it right. The phases agree; the descriptions do not; and *no single distribution over the shared territory restricts to both*.

Why the framework needs it. A theory that identifies unity with synchrony cannot mark that difference, and so cannot separate a brain whose regions oscillate together from one whose regions are about the same world. If the unity of experience is a matter of local contents being about one world rather than merely co-timed, then compatibility, not coherence, is the property

to look for—with coherence entering as the physical condition under which compatibility can be *maintained*, not as a substitute for it.

The statement. The development separates two inputs to the gluing theorem, and only one of them is discharged by dynamics.

Thermodynamic equilibrium (discharged).

The phase field sits at the minimum of the coupling potential. This *is* supplied by dynamics on a class of initial data—the convergence theorem of Section 15 builds the required structure, and the witness is a three-patch cover whose phase field is the limit of a trajectory started *unsynchronized*.

Overlap agreement (not discharged).

Patches at a common phase agree where they overlap. **No dynamics in the development derives this.** It is a class field of the cover the theorem is handed, so it is assumed alongside the eight edges rather than carried by one of them.

Overlap agreement has a concrete pointwise reading: on a discrete substrate, two patch sections have the same restriction to their overlap exactly when they assign the same mass to every site they share (`restrict_eq_iff_density0n_eq0n`). And it genuinely excludes data the rest of the structure admits—the development carries a **counterexample** rather than merely noting the possibility:

A two-patch cover of a three-site substrate, all patches at *one common phase*, whose local density profiles differ at the shared site. It satisfies every other requirement of the gluing theorem and fails compatibility alone (`overlap_agreement_fails`).

That object is what a prose table of assumptions cannot carry. It is also the target the compatibility observable of Part XV would have to separate from its compatible counterpart.

What it does not give. The theorem does *not* require patches to agree away from their overlaps—the three-patch witness carries three pairwise-different profiles. And the whole construction is a formalized gluing *mechanism* under an explicit compatibility postulate, not a derivation of compatibility from synchronization. Neither field continuity nor synchronization alone guarantees a globally compatible representation.

In Lean. `Phase5_GlobalSection.lean`: `probability_glue_unique`, `ThermodynamicCover`, `LocalSectionSynchronization.to_section.eq`; the equilibrium discharge in `Phase5_EquilibriumBridge.lean` (`ThermodynamicCover.ofConvergence`).

30 Where uniqueness comes from

What it is. Accounts of conscious unity need something that picks *one* experience from among the candidates a system could support. Integrated information theory postulates an *exclusion* principle for this purpose (Tononi et al., 2016).

The statement. Here uniqueness arrives twice as a *consequence* rather than a postulate: given a cover whose local sections agree, the sheaf condition gives exactly one global section; and given a contracting self-prediction map, Banach’s theorem (Part XII) gives exactly one fixed point.

What it does not give. The manuscript states the limit of this exactly, because a stronger reading is available and is not theirs. **Uniqueness is relative to a cover and a map.** Choosing them—which regions, which overlaps, which encoding region, and hence where the subject ends—is supplied by neither theorem; that is what E78 and E89 carry. What the construction removes is the need for an additional principle to select among the experiences a

fixed cover and map would otherwise admit. It does not remove the need to say which cover and map a given subject has.

Part XII. Topological obstructions in the phase field

31 Twisted gluing, and why it fails to add content

What it is. A natural weakening of exact overlap agreement: allow local states to differ by a *phase* on each overlap, and ask when the family still glues after one rotation per patch. In sheaf-theoretic language this is asking whether the family of offsets is a *coboundary*—whether the offset on $U_i \cap U_j$ can be written as $\phi_i - \phi_j$ for some per-patch phases.

The statement. Both directions are proved: coboundary offsets glue after one rotation per patch, and conversely, when the phase action is free on overlaps, a family that glues up to phase has coboundary offsets.

And that is precisely why the construction adds nothing here. Any Kuramoto configuration—including a twisted or splay state—supplies absolute patch phases ϕ_i , so its offsets $\phi_i - \phi_j$ are *automatically* a coboundary, and the twisted family always glues. Phase frustration generates no obstruction by this route.

What it does not give. The formal quotient is deliberately *not* called Čech cohomology, because its total cochains still assign irrelevant values to patch pairs with empty overlap. Naming it after the real thing would overstate what is built.

In Lean. `Phase5_TwistedGluing.lean`: `gluesUpToPhase_of_isCoboundary`, `isCoboundary_of_gluesUpToPhase`, `isCoboundary_of_phaseField`, `PhaseObstruction`.

32 Phase singularities and winding numbers

What it is. A narrower and genuinely positive answer comes from the data. The *analytic signal* $z = Ae^{i\phi}$ assigns an amplitude and a phase to a recording at each point. Around a point where the amplitude vanishes—a *phase singularity*, the centre of a spiral or rotating wave—the circle-valued phase need not admit a single real-valued *lift*. Local lifts exist on small patches, and on overlaps they differ by integer multiples of 2π . Summing those integer transitions around a closed loop of electrodes gives the *winding number*, an integer that counts how many times the phase wraps.

This is measured, not stipulated: Townsend and colleagues (2015) identify phase singularities in multielectrode primate LFP recordings by an integer winding number around electrode loops, with the associated complex waves modulating spiking; Xu and colleagues (2023) find task-dependent spiral centres and rotation directions in human fMRI phase maps.

The statement. The formalization treats the finite electrode loop: `loopWinding` sums its integer transitions, `loopWinding_eq_zero_of_hasGlobalLift` proves that transitions arising from a single global lift telescope to zero—so non-zero winding *obstructs* a global lift—and a winding-one witness on three electrodes shows the obstruction is realized.

The cover-level specification is explicit: cortical surface M , analytic signal $z = Ae^{i\phi}$, singularity set Z_t , punctured domain $M_t^* = M \setminus Z_t$, and measurement map $q_t = z/|z| : M_t^* \rightarrow S^1$. A trial-independent electrode or source-space cover of M_t^* supplies local real lifts; their differences divided by 2π are integer transitions on connected overlaps; and loops in the nerve of the cover are the inputs to `loopWinding`.

What it does not give. Two limits, stated firmly. First, this does *not* rescue the twisted-gluing quotient: that quotient uses circle-valued differences of absolute patch phases, while the measured obstruction uses integer transitions between local lifts. They are different objects. Second, **no theorem connects winding sectors to content.** A proposed content-sensitive test is specified—nested preregistered classifiers, a control model containing stimulus, task, behavioural response, eye movement and arousal, an augmented model adding signed winding, source-localized singularity position, rotation direction and lifetime, evaluated on held-out participants and sessions with label permutations within strata, requiring replicated incremental out-of-sample discrimination including under a changed or absent motor report. A positive result would be an empirical *association*, not an identification of a topological sector with phenomenal content; a null increment rejects the proposed role. In finite-resolution data the amplitude threshold defining Z_t , the loop geometry and the required cross-scale persistence are fixed using training or baseline data, never the content labels of test trials.

The construction and its test sit *outside* the conditional chain. A full nerve-indexed integer complex is deferred until a measurement protocol warrants it.

In Lean. `Phase5.PhaseLifts.lean`.

Part XIII. Metric spaces, contractions, and the reflexive self

33 Banach’s fixed-point theorem

What it is. A *metric space* is a set with a notion of distance. It is *complete* when every Cauchy sequence converges—no limits are missing. A map f is a *contraction* with rate $\lambda < 1$ if it shrinks all distances by at least that factor: $d(f(x), f(y)) \leq \lambda d(x, y)$.

Banach’s fixed-point theorem (Banach, 1922). A contraction on a non-empty complete metric space has exactly one fixed point, and iterating the map from any starting point converges to it.

The proof is three lines: iterates form a Cauchy sequence because distances shrink geometrically; completeness supplies a limit; continuity makes the limit a fixed point; and two fixed points would have to be at distance $\leq \lambda d$ from each other, forcing $d = 0$.

Why the framework needs it. A unified global state is not yet a *self*. The framework’s proposal is that a self is a state that models itself accurately: take the map sending a global state to what it predicts it will become, and ask for a state that is its own prediction. If the map contracts, Banach gives exactly one such state—so the self is a *consequence of the construction*, not an additional posit.

The statement.

Theorem (Conditional auto-resonance). On a complete metric space of global sections, a self-prediction map with Lipschitz rate

$$\lambda = e^{-(K-K_c)\tau/2} < 1$$

has a unique fixed point.

The rate in the exponent, $(K - K_c)/2$, is the growth rate of the incoherent state’s instability, so supercriticality $K > K_c$ makes the factor less than one. Relaxation about the *coherent* branch

is twice as fast—linearizing $\dot{r} = \frac{1}{2}(K - 2D)r - cr^3$ about its stable fixed point gives $(K - K_c)$ —so λ is the larger of the two factors and the Lipschitz obligation stated with it is the weaker. Both are below one on exactly the half-plane $K > K_c$. Two results split the parameter plane at exactly Sakaguchi’s threshold, and the negative half is proved as well as the positive: at or below threshold, contraction at that rate is false for *every* self-map of *every* metric space, so the argument is *unavailable* rather than merely unproved.

What it does not give. Coherence supplies $K > K_c$; it does not supply the Lipschitz law. The bound is an obligation on each instance, not a consequence of field dynamics, and no field dynamics in the development produces the read-out map. E89 carries both the Lipschitz law and the identification of the glued section with the fixed point—Banach supplies uniqueness, not that identification.

34 The metric on global sections, and why the obvious ones fail

What it is. To apply Banach one needs an actual complete metric on the space of global sections. Two natural candidates fail, and the development records both failures.

The statement. *The discrete (0/1) metric fails informatively.* Under it, every contraction is constant (`contracting_implies_const`)—because distances take only the values 0 and 1, and shrinking 1 by any factor < 1 forces it to 0. So that metric supports no informative fixed point.

The Lévy–Prokhorov metric—named in the file header—is not what is constructed, and the supplement says why. On a discrete substrate the ε -thickening of a set is the set itself for $\varepsilon < 1$, so the averaging map $\mu \mapsto \frac{1}{2}\mu + \frac{1}{2}\mu_0$ is not a Lévy–Prokhorov contraction there; and Mathlib carries no completeness result for that metric anyway.

What is constructed is the uniform distance between densities: the largest difference between the masses two sections’ measures put on any single site. It is proved complete by exhibiting the limit section of a Cauchy sequence of densities, and distinct sections at distance $1/2$ are exhibited so the space is not degenerate. On a finite substrate this is equivalent to total variation of the glued measures within a factor of the number of sites—but *not equal* to it, and no claim of equivalence with Lévy–Prokhorov is made.

34.1 The germ–measure dictionary

Getting from sheaf sections to numbers requires a bridge, and Mathlib supplies none: `TopCat.Presheaf.sheafify` has no adjunction and no `isIso.toSheafify`, so the inverse had to be built by hand.

On a *discrete* substrate every point has a smallest open neighbourhood—itsself—so “the mass carried at x ” is a well-defined map out of the stalk at x , constructed as the colimit map of a cocone whose cocone law is precisely that restriction preserves point masses. Reading a section’s germ at each site of an open U gives a map to mass profiles on U , and `massEquivOn` proves it a *bijection*: injective because a germ on a discrete space is determined by its restriction to the point, surjective because every prescribed profile is realized by a finite measure on U .

35 The avatar: making the self-model run through a region

What it is. The self-prediction map could have been carried as free data sitting beside the structure. The development declines to, for a reason worth understanding.

The statement. If `predict` were a free field, then a *blinded* boundary—one whose self-encoding region is replaced by a constant—would retain the same predictive model and the same fixed point, since nothing would tie the map to the region. So the structure instead carries the region’s *write* (`auto_resonance`) and its *read-out* (`readout`), and `predict` is their *composite* rather than a field. The self-model runs through the encoding region—the “avatar”—or it does not exist.

Four theorems fix the status of the resonance condition, which is a *predicate* (`IsRestrictionResonance`: the write is the presheaf restriction of the state to the avatar region) rather than a field:

- `eq_of_avatar_eq`: if the avatar regions of a family of resonant boundaries *cover* the substrate, two global states with the same reading at every avatar are equal. The readings are local; the global state is reconstructed from them by the same sheaf condition Part XI uses.
- `self_mem_range_readout`: a fixed point lies in the range of the read-out, so the Self is reconstructed from a single section over the avatar region—the *low-dimensional fold*, as a theorem.
- `self_eq_readout_restrict`: under resonance the Self satisfies $s = \text{readout}(s|_{\text{avatar}})$ —the field is what its own localized self-encoding reconstructs. This is exactly the statement a structure carrying `predict` as free data could not make.
- `not_self_of_constResonance`: the negative half. The blinded boundary’s map is constant, its only fixed point is named, and the original Self is *not* one. Note the asymmetry: the opposite statement—that blinding leaves the Self where it was—does not even typecheck, since a contraction hypothesis about one boundary’s map says nothing about another’s.

What it does not give. Three limitations are stated in the source and should be repeated here. (i) The read-out is a *modelling choice*: no field dynamics produces it, and the Lipschitz bound at the mean-field rate is an instance obligation. (ii) The witness substrate’s coupling value is a choice of units, not a measurement. (iii) Nothing derives the resonance condition—that a physical avatar performs the restriction is an assumption on an instance, in the same class as the overlap-agreement obligation of Part XI.

In Lean. `Phase6ReflexiveTopology.lean`, with the three-site witness in `Examples.lean` §10 exercising every clause—including `cortexBlind_self_ne` (the baseline is *not* a fixed point of the blinded map) and `cortexBlind_not_determined` (two distinct states with identical blinded readings).

Part XIV. Coupling support, atomless measures, and hardware

Two formal comparisons bear on the substrate question. Both admit a stronger reading than they support, and the manuscript spends more words disowning the stronger reading than stating the results.

36 Resource-matched coupling reallocation

What it is. *Rigidity* here means a constraint on *which pairs can be wired at all*—weights tunable, wiring fixed—not a constraint on weight values. The predicate `RealizableIn S R A` says A is a valid coupling meeting resource budget R and vanishing on every pair the architecture has no wire for.

The statement. If phase correlation over the *wired* pairs is bounded by M , then no realizable coupling achieves total correlation above $R \cdot M$, and that bound is attained. Consequently, if some pair the wiring *misses* has correlation exceeding M , every realizable coupling is strictly suboptimal—and the quantitative gap is at least $R(\cos(\theta_l - \theta_k) - M) > 0$ at equal budget. Note that this *derives* the suboptimality hypothesis that the reallocation lemma would otherwise have to assume: it is a consequence of the wiring pattern, not an extra premise.

What it does not give. The result is resource-matched and not question-begging, but it turns on *fixed wiring support*, not on continuity. **An architecture wired to the best pair is not beaten.** It is a statement about constrained optimization, not about biology.

37 The continuum/discrete separation

The statement. `fieldCorrelation` integrates a kernel against $\mu \otimes \mu$. A finitely-sited architecture is a finite set of points; on an *atomless* substrate such a kernel contributes exactly zero to the coupling energy, for every phase field and every choice of weights, while a patch-supported kernel gives $c\mu(U)^2$. So a continuum observable cannot register a finitely-sited architecture at all.

What it does not give. This one is a genuine type-level separation, but it is **not resource-matched**—a measure-zero substrate carries zero resource in the $\mu \otimes \mu$ sense as well. What it establishes is a claim about the *model’s observable*, not about computational power. And as Section 27 noted, it cuts against the framework’s own coarse-graining step as much as against anything else.

Neither result formalizes manifold structure or lattice geometry. Neither establishes that digital hardware cannot approximate field dynamics or support consciousness. No biological-versus-artificial conclusion follows from either. The manuscript’s identification is, as stated, multiply realizable: nothing in the sheaf or fixed-point conditions is specific to cortex or to electromagnetic coupling, so the substrate claim is carried entirely by E56 and E67 and is empirical.

In Lean. `Phase7.Rigidity.lean: totalCorrelation_le_of_realizable, exists_realizable_pair, rigid_is_strictly_suboptimal, rigid_gap, exists_better_coupling_allocation; fieldCorrelation_sited.eq_zero fieldCorrelation_patchKernel, sited_architecture_below_field_optimum.`

Part XV. Formal method: why the development declares no axioms

This part contains no physics. It is here because the manuscript’s fourth contribution is a claim about method, and that claim rests on one technical point.

38 What formalization is claimed to add

What it is. A prose table of assumptions would list the same eight edges. What the Lean development adds is that it makes the list *non-negotiable*.

The statement. The characteristic failure in an argument of this shape is not a false step but a *silent change of subject*:

- a theorem about a stationary density read as a claim about a trajectory;
- an inequality about a register read as a claim about learning;
- agreement of phases read as compatibility of representations.

Type-checking forbids each of those substitutions. An assumption needed to make a step go through appears as a hypothesis *in the term*, rather than as a word in a sentence. The last substitution is not merely forbidden but *refuted by a constructed object*—the incompatible-at-equal-phase cover of Section 29—and that object is what a prose table cannot carry.

What it does not give. Formalization does not make the assumptions true. It makes them impossible to lose.

39 Why a global axiom is the wrong shape for a physical postulate

What it is. Formalization is only as trustworthy as its axioms, and an *inconsistent* axiom silently makes every theorem downstream of it vacuous while leaving the build green. The failure mode is specific and easy to reproduce.

The statement. The failure has one shape: **an axiom that constrains a symbol it does not itself bind**. Such a declaration is not a postulate about one system; it is a constraint on *every* model of the class the symbol belongs to, including models built to violate it—and a class with such a model is thereby inconsistent. Three concrete cases, each of which proves **False** on its own:

1. A symmetry-breaking statement taking the potential-energy functional as an *implicit* argument occurring only in its hypothesis. Instantiating that functional with the constant map discharges the hypothesis by `rf1` and yields the conclusion for an arbitrary field. With $V(v) = v^2$, $v_0 = 0$ and $\phi \equiv 1$ this proves $1 = 0$.
2. A Landauer heat statement equating a free class field with a bath-entropy difference *for every instance*. An instance with zero heat dissipation and a bath gaining one bit gives $0 = \log 2$.
3. A KL bound asserting $\sigma \geq D_{KL}(P\|Q)/\Delta t$ for *all* P, Q , while σ is a single real fixed by a class field. Since D_{KL} is unbounded above, a point mass against a uniform distribution refutes it.

Two further declarations of the same shape are recorded because their refutation is different in kind: as global assertions pinning free fields, a phase-periodicity condition and a synchronization-to-section equation are refutable as soon as the probability presheaf admits two distinct global sections. That refutation needs no arithmetic at all, only that the sheaf is non-trivial.

The rule that prevents all of this is uniform:

*A physical postulate that mentions a class field must be a **field of that class**, never a standalone axiom quantified over all instances.*

A postulate is then an *obligation each model discharges*, not a global claim about every model. Under that rule the development declares no axioms at all: `#print axioms` on any result reports only Lean’s own foundational three—`propext`, `Classical.choice` and `Quot.sound`.

40 Witnesses, and what inhabitation does and does not prove

What it is. A conditional theorem about an *uninhabited* class is vacuously true and worthless. So every class carrying a physical postulate gets an explicit witness in `Examples.lean`: a one-bit erasure system with a genuine bath reproducing $\Delta Q = k_B T \ln 2$; a perfectly resonant system discharging the KL bound; a scalar field on a one-point spacetime discharging the action principles; and—the most demanding case—a thermodynamic cover over a three-site substrate with the discrete topology and its Borel σ -algebra, covered by two overlapping open patches meeting at a single shared site.

What it does not give. What the witnesses establish is that the classes are **inhabitable**, so the theorems about them are not vacuous. What they do not establish is **that a cortex is such an instance**. In the cover above the phase field is already constant—every patch locked at $\theta = 0$ —which is precisely what makes the equilibrium obligation dischargeable. Getting a physical system into that configuration is the content of the informal dynamical argument, not of the witness.

41 The chain theorem

The statement. `Chain.lean`'s theorem `chain` takes the eight named hypotheses E12–E89, together with a `ThermodynamicCover` and a `ReflexiveBoundary` whose class fields carry physical content of their own, and concludes `UnifiedSelf`: that the particular section glued by a thermodynamic cover *is* the unique fixed point of self-prediction. `chain_hypotheses_jointly_satisfiable` discharges all eight at once on a witness assembled from a double well, a register, a uniform-grid mesh and a three-site cortex, which remain distinct toy systems rather than one mechanism.

Two of the eight ask for more than bare existence, and each has a regression test rejecting a structure that would satisfy the weaker form:

- **E34** asks for a predictive structure carrying *the register's own* temperature and Landauer heat, with non-zero waste—not merely for some predictive structure.
- **E78** asks for a cover coupled at least at K whose equilibrium is *the limit of a relaxation on that coupling* (`ThermodynamicCover.IsReachedByRelaxation`)—not merely for some cover.

What it does not give. The joint witness proves the hypotheses *simultaneously consistent*. It does not prove that one mechanism realizes them, and it certainly does not establish the empirical truth of any premise. E12 remains equivalent to its own conclusion, which is why `n1` and `n2` are described as independent roots rather than as a derivation.

Part XVI. Circular statistics and the empirical test

42 Doing statistics on angles

What it is. Phases live on a circle, where ordinary statistics break. The mean of 1° and 359° is 180° if you average the numbers and 0° if you average the directions; only the second is right. Circular statistics works with the unit vectors $e^{i\theta_j}$ instead:

Resultant length r .

$r = \left| \frac{1}{N} \sum_j e^{i\theta_j} \right|$. Zero for a uniform spread, one for perfect alignment. This is the order parameter.

Circular mean ψ .

The direction of that same resultant vector.

Concentration a .

The von Mises parameter, governing how tightly the distribution is peaked.

Why the framework needs it. Because the manuscript's protocol turns on estimating a and r *independently*, and that is the constraint most easily violated in practice: linear statistics on wrapped phases estimate neither quantity.

43 Getting the phases: the analytic signal

What it is. The *Hilbert transform* turns a real oscillatory signal $x(t)$ into a complex *analytic signal* $z(t) = x(t) + i\mathcal{H}[x](t) = A(t)e^{i\phi(t)}$, from which an instantaneous amplitude and an instantaneous phase can be read. It is meaningful only for reasonably narrowband signals, which is why the pipeline bandpass-filters first.

The statement. The implemented pipeline: detrend each channel; bandpass with a 4th-order Butterworth filter in the 4–40 Hz band using second-order sections (`sosfiltfilt`, which is zero-phase); take the Hilbert transform for $\theta_j(t)$; use only the 62 scalp EEG channels, discarding EOG and EMG. Three seconds of padding on each side is downloaded and trimmed after filtering to suppress edge artifacts.

44 The parameter-free prediction, and the trap in testing it

What it is. The stationary model predicts that concentration and coherence lie on one curve:

$$r = \frac{I_1(a)}{I_0(a)},$$

with *no fitted shape parameter*—the same curve for every subject and every state.

The statement. The estimator must not be a maximum-likelihood one, and this is the sharpest methodological point in the empirical section. The von Mises family is a one-parameter exponential family whose sufficient statistic is $\sum_j \cos \theta_j$ —that is, r itself. So *any* ML estimate of the concentration solves its own score equation $I_1(\hat{a})/I_0(\hat{a}) = r$ and returns $\hat{a} = A^{-1}(r)$ identically. The pair $(\hat{a}, r)$ would then lie on the predicted curve *for any input whatsoever*, including phases drawn from a distribution that is not von Mises, and the collapse would be an algebraic identity rather than a test.

The implemented estimator therefore uses the log-density *shape*: phases are centred on their circular mean, binned into 40 bins on $[-\pi, \pi]$, and the log of the binned density (with a 0.5 pseudocount) is regressed on $\cos \theta$ by *unweighted* ordinary least squares. Under the von Mises model $\log \rho(\theta) = a \cos \theta - \log(2\pi I_0(a))$, so the slope is a . Other weighting rules are different estimators and are not assumed equivalent to maximum likelihood.

Discriminating power was verified directly. At the analysis’s sample size, a wrapped Cauchy ($\rho = 0.6$), a top-hat arc and a 50/50 von Mises–uniform mixture give residuals $r - I_1(\hat{a})/I_0(\hat{a})$ of +0.088, −0.117 and +0.075, against a worst case of 0.0024 on genuine von Mises samples—a separation of $31.7\times$. A uniform distribution passes, correctly, being von Mises at $a = 0$. The pseudocount biases the estimator by ≤ 0.007 for $a \leq 3$, reaching −0.375 only at $a = 5$ where the troughs empty out.

What it does not give. Even a perfect result here identifies neither ephaptic coupling nor this framework: the relation is common to the whole von Mises family, so agreement with it does not distinguish this framework from alternative synchronization models. Worse, within the same ansatz $a/r = K/D$ for $r > 0$, so the limiting threshold ratio *follows from* the density relation instead of testing it—and ratios near the finite-size floor are especially unstable.

45 What the EEG analysis found

The statement. On public propofol-sedation EEG (OpenNeuro ds005620, [Bajwa et al., 2025](#)), 21 subjects at 65 channels and 5 kHz, analysed in non-overlapping 100 ms bins:

- **Cross-subject (8 subjects), bipolar montage.** Mean residual -0.0002 ± 0.0002 (SEM), RMSE 0.0005, max $|r - I_1/I_0(a)| = 0.0013$. Group $a = 0.373$ (range 0.302–0.542), $r = 0.173$.
- **Common-reference montage.** Mean residual -0.0057 ± 0.0028 , RMSE 0.0093, max 0.0246—an order of magnitude worse, and *systematically negative* in every block and every subject.
- **Narrowband.** Theta, alpha, beta, gamma and broadband all collapse (mean residuals −0.0018, +0.0001, +0.0001, +0.0000, +0.0000). The collapse is not band-specific.

- **Window sensitivity.** Stable over 20–100 ms pooling windows; not at 5–10 ms.

The montage gap is the one systematic feature. A common-mode component that simply raised r would push the residual the other way, so that is not the mechanism; the sign is instead consistent with the top-hat synthetic control, suggesting a shared reference imposes a sharp common-mode peak on channels that remain broadly distributed around it, steepening the fitted a more than it raises the pooled resultant length. This is reported as a candidate mechanism whose direction fits, not a demonstrated one.

What it does not give. Four limits, and the third is the one that matters most.

1. **The estimand is not what the protocol asks for.** 62 channels pooled over a 100 ms window estimates a *spatiotemporal* distribution, not an instantaneous spatial one—a departure forced by channel count, and one that any scalp implementation will face. Pooling is required because the independent log-density estimator needs at least 100 phase samples.
2. **Referencing is a material analysis choice,** not a preprocessing detail. Bipolar and common-reference give different answers.
3. **The concentration range is too narrow to be informative about the curve.** Across the eight subjects a spans only 0.302–0.542. Over so short an arc the Bessel ratio is nearly straight: it departs from the naive linear form $a/2$ by at most 0.0094 and from $\tanh(a/2)$ by at most 0.0030, while per-bin residual scatter is 0.02–0.03. **The data are compatible with the relation but do not discriminate I_1/I_0 from a straight line.** Doing so needs recordings reaching $a \gtrsim 1.5$, where the two separate by more than 0.17.
4. **Bootstrap intervals are exploratory.** Resampling time bins (2000 resamples, percentile method) does not preserve dependence between neighbouring bins.

And the purpose was never to test the awakening hypothesis, for which these are the wrong data: they concern propofol sedation, not recovery from sleep. The exercise demonstrates what the protocol produces and where it is fragile.

46 The awakening experiment

The statement. The manuscript specifies six requirements, and the design—not any single measurement—is where the content of the hypothesis lives.

1. **Coupling and diffusion in the same units.** K and D estimated from the same cortical regime, in inverse-time units, separating phase diffusion from quenched frequency spread and from observation noise. Without this, “ $K > 2D$ ” is not an evaluable statement about cortex.
2. **Geometry calibrated to coupling.** A diffusion-MRI observable or astrocytic signal indicates the extracellular volume fraction α ; its mapping to effective coupling must be established separately, with its own uncertainty. The conductivity argument below supplies the expected sign of that mapping and nothing more.
3. **Adequate spatial resolution.** At least 100 spatial samples per estimate, r by resultant length and a by unweighted log-density regression. With fewer sites, pooling over time changes the estimand—and must be stated as such.
4. **An independent recovery endpoint.** Behavioural recovery measured separately from the phase trace, so latency is not defined by the signal whose latency is at issue.
5. **A predeclared onset criterion,** fixed above the finite-size floor before unblinding, chosen knowing that a finite ramp smears the cusp.

6. **Model comparison under matched observation.** Against exponential, flexible sigmoidal and other slow-recovery models under identical temporal filtering and observation noise. Latency alone cannot distinguish the mechanism, because many processes create delays.

The statement. The design fails informatively in three ways: failure of the stationary phase-density relation over a sufficiently broad concentration range; absence of the proposed coupling–geometry association; or a recovery trace better explained by an alternative after finite-rate and observation effects are accounted for.

What it does not give. Two apparent failures are not failures. A linear apparent onset is evidence against the adiabatic approximation only where that approximation is independently justified—see Section 23—and does not falsify the field hypothesis under arbitrary ramp speeds. And the absence of a fold in the stationary equation excludes stationary coexistence of two positive branches in this model, but does not exclude finite-rate hysteresis or adjudicate theories of ignition.

Part XVII. Dimensional analysis and the calibration problem

47 Why the units do not match

What it is. The manuscript calls this “the sharpest open limitation of the framework”, and it is a pure dimensional-analysis problem—the kind that is embarrassing precisely because it cannot be argued away.

The statement. A natural Fermi estimate of field-mediated coupling multiplies a population size N , a spike-time shift per unit field s , a field amplitude E , and an oscillation frequency f . Track the units:

- s is a time per unit field, so sE is a *time*;
- multiplying by a frequency f gives a *pure number*—the fractional phase advance a cell receives per cycle from the population field;
- multiplying by a population size N leaves it a pure number.

But a coupling in the sense of the stochastic model (1) is an *inverse time*, comparable with D . So the estimate cannot be compared with D at all. The conversion

$$K_{\text{eff}} = \gamma N s E f$$

puts the entire gap into a rate calibration γ .

48 Why γ is not recoverable

The statement. The substantive point is that γ is a *hypothesis about persistence*, not a unit conversion to be fixed by convention. It states how a per-cycle phase advance accumulates into a drift rate, which depends on how long a field-driven timing shift survives. Three candidates are available and they are not interchangeable:

- $\gamma = f$: each cycle’s advance is retained in full and independently;
- $\gamma = 1/\tau$ for a phase relaxation time τ : it decays as the medium relaxes;
- $\gamma = D$: it accumulates at the rate phase itself diffuses.

These need not agree, in cortex or anywhere else, and a field measurement chooses between them only if the persistence of induced timing shifts is measured alongside it.

The third must be declined for an additional reason, and the reason is instructive. Under $\gamma = D$ the threshold test $K_{\text{eff}}/D > 2$ reduces to $NsEf > 2$ —a comparison of the dimensionless field estimate against the number 2, *with the diffusion parameter cancelled out of a test whose entire content was the ratio of coupling to diffusion*. The economy of that substitution is what makes it unsafe: it converts an unmeasured rate into an apparent confirmation while removing the quantity the confirmation was about. The same objection applies to any γ recovered by requiring the model to cross its own threshold, or read off the Bessel collapse used to test the stationary density.

What it does not give. Every saved field-strength value in the repository is therefore *conditional* on a calibration convention. Under the numerical convention $\gamma = 1$ in the simulation’s inverse-time unit, the Fermi arithmetic gives required field ranges that fall *below* the 1–5 mV/mm comparison band, and they scale inversely with γ . The simulation does not measure the calibration, so it cannot independently validate physical-unit consistency or a cortical field threshold. Identification requires coupling and diffusion estimated in one cortical regime by a route *independent* of the field arithmetic—which is item 1 of the protocol above.

49 The empirical case for the field, and its critics

The statement. The identification of the continuum kernel with cortex’s endogenous electromagnetic field is supported as *modulatory and integrative*, not as primary. Transmembrane currents generate extracellular fields that shift neighbouring membrane potentials—ephaptic coupling. Endogenous fields of 1–5 mV/mm entrain spike timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015); stable electric fields carry information that survives representational drift in the underlying spiking population (Pinotsis & Miller, 2022, 2023a,b); the field is spatially continuous and superposes contributions from large populations, which is the integrative role CEMI field theory assigns it (McFadden, 2002, 2020; Pockett, 2012), physically supported by astrocytic syncytia joined by gap junctions (Verkhatsky & Nedergaard, 2018). Sleep-associated changes in extracellular space and clearance (Xie et al., 2013) motivate the possible slow control of coupling.

What it does not give. The formal predicate that E56 carries is explicit and modest: a jointly continuous kernel, a normalized substrate, a coupling constant equal to that kernel’s double average, no single site contributing more than half of it, and a noise D that is the field’s own. **No object in the development denotes cortex**, so whether cortex satisfies these is not a Lean question. Toy witnesses satisfy the conditions; none is a model calibrated to cortex.

The longest-standing objection is that the fields are too weak to matter, pressed from *within* the field-theoretic camp by Pockett (2012, 2002). Vöröslakos et al. (2018) found that applied current is shunted through scalp, delivering fields at or below 1 V/m in cortex. That bounds *applied* rather than endogenous fields, but it constrains how much causal work a field of that magnitude can do—and is the reason the framework does not claim the field drives spiking. Changes in phase organization under propofol (Bastos et al., 2021) likewise do not measure a crossing of the model’s coupling threshold.

Extracellular geometry reaches the coupling through the field amplitude and not through a further mechanism. The extracellular space occupies a volume fraction $\alpha \approx 0.2$ of cortical tissue at a diffusion tortuosity $\lambda \approx 1.6$ (Syková & Nicholson, 2008), and natural sleep raises α from roughly 0.14 to 0.23 (Xie et al., 2013). Cortical tissue conducts approximately resistively below 10 kHz, at 0.3–0.6 S/m, an order of magnitude set by the conductivity of saline diluted into that volume fraction (Barbour, 2017). At a fixed transmembrane current density, then, a larger α means a smaller extracellular field: the space contracts on awakening and K should rise. That

is an effective-medium argument, so it fixes a sign and not a magnitude, and it says nothing about how a heterogeneous field aggregates into a single K . Neither the monotonic relation between extracellular geometry and coupling, nor its role in human sleep inertia, follows from the cited physiology; the model requires both to be measured.

The identification would be challenged if independently calibrated field-mediated coupling were insufficient for the proposed regime, or if a selective perturbation of that interaction left the proposed neural and behavioural endpoints unchanged. Such a perturbation needs controls for direct effects on transmembrane currents and other coupling pathways; selectivity is not assumed experimentally available.

Part XVIII. What the mathematics is asked to mean

The composition terminates in a mathematical object, and the manuscript takes a position on what that object is. This part states the position, the three arguments for it, its costs, and the missing observable that currently keeps it from being tested.

50 The commitment

The statement. Two clauses, neither a theorem:

Unity.

The unique global section glued from compatible local states is the formal correlate of the unity of a conscious episode—the fact that a subject undergoes *one* experience rather than a collection of co-occurring local ones.

Selfhood.

The fixed point of the self-prediction map on that same section is the formal correlate of a minimal self—a state that is what it takes itself to be.

51 Three arguments, and what kind of arguments they are

1. **Compatibility rather than agreement.** Phases can agree everywhere while local probabilistic descriptions remain incompatible on their overlaps, and Section 29 exhibits that configuration *inside the formalization*, not merely as an illustration. If unity is a matter of local contents being about one world rather than merely co-timed, compatibility is the property to look for.
2. **Uniqueness as consequence, not principle.** As Part XI noted, uniqueness arrives twice from the construction rather than from a separate exclusion postulate—though only relative to a given cover and map. This is *a structural argument, not evidence*.
3. **The self as condition rather than addition.** The predictive map restricts a global state to a region encoding it and reads that encoding back out, so a fixed point is a state whose self-model is accurate about the state it models. That condition is *available at all* only because the state modelled is a single glued object: a system whose local descriptions do not cohere has no such state for a self-model to be about. Unity and selfhood are one construction read at two levels—which is why the chain's conclusion requires the fixed point to *be* the glued section, rather than some unrelated section.

52 The costs

What it does not give. The manuscript states these rather than leaving them to a referee.

- **It does not address the hard problem.** The commitment is a proposal about *which* physical state is the conscious one, not about why states of that kind feel like anything. It is therefore open to the objection that a system could satisfy every condition—compatible local states, a supercritical field, an accurate self-model—while nothing is experienced. *No reply to that objection is offered or claimed.*
- **It is multiply realizable as stated.** Nothing in the sheaf or fixed-point conditions is specific to cortex or to electromagnetic coupling, so the substrate claim rests entirely on the two empirical edges E56 and E67. No conclusion about artificial systems follows from the formal results.
- **It is independent of the physical tests.** The predictions of Parts VIII, IX and XVI concern coupling, coherence and recovery. They can fail while the identification stands, or succeed while it is wrong.

53 What would count against it, and the observable that is missing

The statement. The commitment predicts that the two clauses and the presence of a unified conscious episode go together. So *dissociation* would tell against it: a state in which local cortical descriptions are demonstrably compatible and the field supercritical, while the subject reports no unified experience—or unified reported experience where compatibility demonstrably fails. Anaesthetic emergence, dreaming and dissociative states are the places to look, because they separate coherence from report in ways ordinary waking does not.

Such tests cannot presently be run, because coherence has an observable—the resultant length—and *compatibility does not*. The supplement therefore specifies what one would have to satisfy. It is a specification, not a result: nothing is implemented, and the construction is left to separate work.

53.1 Three components

A restriction map.

Two regions with overlapping receptive territories each carry a distribution over what occupies the shared part, decoded from that region’s activity alone. Compatibility is a relation between the two *restrictions*, so the map taking a region’s decoded distribution to a distribution over the overlap must be specified before any statistic is computed. On a discrete substrate this is the marginalization the formalization already uses.

A statistic with a null.

A distance between the two restricted distributions is not yet a measurement, because decoding error alone makes that distance positive. The statistic needs the distribution of the distance *under compatibility*—what two decoders of the same underlying content produce given their own noise—so that incompatibility is a departure from a stated null rather than a non-zero number. Overlapping territories supply dependent samples, and the uncertainty must be computed accordingly.

Validation where the answer is built rather than inferred.

The estimator must be exercised on data in which compatible and incompatible overlaps are constructed by design. The formalization supplies the configuration to target:

local states at a single common phase whose densities disagree where the patches meet (Section 29). An estimator that cannot separate that configuration from its compatible counterpart at realistic decoding quality is not yet a measurement of the property the framework needs.

53.2 Three ways it fails silently

1. A decoder that shrinks both regional posteriors toward a shared prior reports *compatibility* whatever the regions encode—having replaced two local descriptions with one global one.
2. A decoder carrying independent per-region bias reports *incompatibility* whatever the regions encode.
3. **Circularity.** A compatibility statistic computed from phases measures phase agreement—and the distinction between phase agreement and overlap compatibility is the entire content of E78. An observable that read compatibility off synchrony would confirm the framework’s central distinction by assuming its negation.

The first two are opposite failures of the same component, neither visible in the statistic itself, which is why the constructed-answer validation is a requirement and not a refinement.

What it does not give. Such an observable would *not* establish E78, which asserts that coherence drives a cover the theorem is given to its relaxed equilibrium, nor the cover’s own compatibility field. It would make the consequent measurable and leave the implication a hypothesis. It bears on neither the recovery prediction nor the identification of the glued section with experience. What it supplies is the missing term in the dissociation test. A null outcome—no reachable data separating compatible from incompatible overlaps at achievable decoding quality—is itself informative, and would place the interpretive commitment outside experimental reach rather than merely outside the article.

54 Relation to other accounts

The statement. The proposal differs in emphasis from integrated information (Tononi et al., 2016), global workspace (Baars, 1997; Dehaene et al., 2014), and predictive processing (Friston, 2010). It specifies a candidate interaction medium and a mathematical compatibility condition, and—relative to a given cover and self-prediction map—it *derives* the uniqueness those accounts postulate.

What it does not give. It does not derive the exclusion principle of integrated information, identify every broadcast with a field transition, or derive a generative model from a dissipation bound (Section 9). And continuity alone does not settle the combination problem: continuous fields can carry incompatible local descriptions, and discrete models can admit sheaf constructions. Compatibility is what this account offers in place of continuity. These mathematical differences do not settle which systems have conscious experience.

Appendices

A Master table: concept, role, and status

Read the status column strictly. “Theorem” means proved under stated hypotheses; it does not mean physically established. “Instance postulate” means a physical assumption carried as an obligation each model discharges, with an explicit witness showing the class is inhabitable.

Concept	Role in the argument	Status
Spontaneous symmetry breaking (§3)	Degenerate minima make a choice necessary	Theorem (holds a.e.; the a.e. is necessary)
Topological defects, π_0 (§4)	Forces a boundary for an agent	Theorem; only the domain-wall case taken
Finite capacity	Bounded register cannot absorb its history	Theorem; finiteness assumed and fenced
Landauer’s principle (§6)	Erasure costs heat	Theorem + instance postulate ($\Delta Q = T\Delta S_{\text{bath}}$)
Norton–Shenker distinction	Logical $\neq$ physical erasure	Built into the definitions; survives as a theorem
Data processing inequality	Splits memory into predictive and not	Theorem; no-back-action is load-bearing
Still et al. bound	Nonpredictive memory is costly	Instance postulate; consequences derived
Landauer $\rightarrow$ Still bridge	One heat, two vocabularies	Theorem under an identification (= E34)
KL bound; FEP critique (§9)	Blocks “dissipation $\Rightarrow$ true model”	Instance postulate; the limit does <i>not</i> follow
Squared-drift gradient (§10)	A descendable objective for plasticity	Theorem at fixed phases; <i>not</i> entropy production
Kuramoto SDE	The physical core	Model
Lyapunov descent (§12)	Motion is gradient flow on a potential	Theorem; minimum characterization needs positivity
Rotating frame	Repairs the unbounded potential	Theorem; exact only for identical frequencies
Picard–Lindelöf, global Barbalat’s lemma	Trajectories exist and are unique Motion stops	Theorem on all of $\mathbb{R}$ Proved here (absent from Mathlib)
Łojasiewicz estimate (§15)	Locates the limit	Theorem, conditional on initial data
Propagation of chaos (§16)	Would link finite N to mean field	Open ; not closed by any theorem
von Mises density (§17)	The stationary phase law	Assumed ; checked numerically only
Threshold $K_c = 2D$ (§18)	Coupling gate for coherence	Theorem about a stationary density
Fold-and-cross uniqueness (§19)	Exactly one coherent branch	Theorem; no differentiation used
Critical exponent $\beta = 1/2$ (§20)	Square-root onset	Theorem; filter-indexed, with witness
Ramp-speed dependence (§23)	Apparent exponent tracks the drive	Simulation; a result about method
Squared-drift plasticity (§25)	Dissipation descent $\neq$ learning	Simulation; negative result
Mesh refinement	Discrete energy $\rightarrow$ continuum energy	Theorem (scalar energy only)
No kernel from coarse-graining (§27)	Continuum kernel is an input	Theorem (negative)
Sheaf gluing (§28)	Local $\rightarrow$ unique global state	Theorem + instance postulate
Compatibility $\neq$ phase agreement (§29)	The framework’s central distinction	Theorem, with explicit counterexample
Twisted gluing	Weakening to per-patch phases	Theorem; adds nothing (offsets are coboundaries)
Winding numbers, phase lifts	Measured topological obstruction	Theorem + empirical candidate; no link to content
Banach fixed point	The reflexive self	Conditional theorem; rate is an obligation

Concept	Role in the argument	Status
Avatar / restriction resonance	Self-model runs through a region	Theorems, positive and negative halves
Rigid wiring suboptimality	Support constraints cost correlation	Theorem; <i>not</i> about biology
Atomless / sited separation	Continuum observable ignores finite sites	Theorem; not resource-matched
No-axioms methodology (§39)	Postulates as instance obligations	Methodological; three refutations recorded
Chain theorem	Eight hypotheses $\Rightarrow$ UnifiedSelf	Conditional theorem; jointly satisfiable
(a, r) collapse (§44)	Parameter-free distributional test	Compatible with EEG; range too narrow to discriminate
Sleep-inertia shape	The discriminating prediction	Untested
Rate calibration γ (§47)	Converts field estimate to coupling	Not derivable ; the sharpest open limitation
Compatibility observable	Needed for dissociation tests	Specified, not implemented
Unity / selfhood identification	What the object is taken to be	Philosophical commitment , argued not proved

B Glossary

Adiabatic tracking

The assumption that a slowly driven system stays on its stationary branch. Fails near a transition, where relaxation times diverge.

Analytic signal

The complex signal $Ae^{i\phi}$ built from a real signal by the Hilbert transform, supplying instantaneous amplitude and phase.

Atomless measure

A measure giving zero mass to every single point. Lebesgue measure is the example; finitely-sited kernels are invisible to integrals against it.

Barbalat's lemma

If a function is well-behaved and its integral converges, the function tends to zero. Not in Mathlib; proved in the development.

Bessel ratio $R(a)$

$I_1(a)/I_0(a)$, the mean of $\cos \theta$ under a von Mises law at concentration a . Strictly increasing.

Coboundary

A family of offsets on overlaps expressible as differences $\phi_i - \phi_j$ of per-patch values. Kuramoto phases always give one, which is why twisted gluing adds nothing.

Concentration a

The von Mises peakedness parameter, estimated here from the slope of the log-density on $\cos \theta$.

Contraction

A map shrinking all distances by a factor $\lambda < 1$.

Critical exponent β

The power law $r \sim (K - K_c)^\beta$ near threshold. Mean-field value $1/2$.

Critical slowing down

Divergence of relaxation time at a continuous transition; what limits adiabatic tracking.

Data processing inequality

Processing cannot create information: $I(X; Z) \leq I(X; Y)$ for a Markov chain $X \rightarrow Y \rightarrow Z$.

Dale’s law

The constraint that a neuron is either excitatory or inhibitory for all its targets.

Euler–Maruyama

The basic numerical scheme for SDEs; noise scales as $\sqrt{\Delta t}$.

Fokker–Planck equation

The PDE governing the evolution of a probability density under an SDE. Its stationary solution is the von Mises density—a step verified numerically, not formally.

Frustration

Signed couplings that cannot all be satisfied at once, producing a near-degenerate landscape. Excluded from the chain by the positivity hypothesis.

Germ, stalk

Data “arbitrarily near” a point, obtained as a colimit over its neighbourhoods.

Gluing axiom

The sheaf condition: compatible local sections assemble uniquely into one global section.

Global section

An element of $\mathcal{F}(X)$ over the whole space, restricting to the given local data.

Kibble–Zurek mechanism

The physics of driven transitions: the faster the drive, the later the escape, with a power-law delay.

Landauer’s principle

Erasing information costs at least $k_B T \ln 2$ per bit.

Lipschitz

Bounded rate of change: $|f(x) - f(y)| \leq L|x - y|$.

Łojasiewicz estimate

An inequality bounding a functional’s excess above its minimum by its current rate of decrease. Yields exponential convergence without compactness.

Lyapunov function

A quantity that never increases along trajectories.

Mean field

The approximation replacing all pairwise interactions by an averaged influence. Justifying it from finite N is propagation of chaos.

Order parameter r

Circular resultant length, $|\frac{1}{N} \sum_j e^{i\theta_j}|$; zero for incoherence, one for full locking.

Pitchfork bifurcation

A transition where a branch of solutions emerges continuously from zero as a parameter crosses threshold, the old solution persisting.

Preconnected / connected

Topological notions of “in one piece”; continuous images of connected sets are connected—the whole proof of the domain-wall theorem.

Presheaf / sheaf

Data assigned to open sets with restriction maps; a sheaf additionally satisfies the gluing axiom.

Propagation of chaos

Asymptotic independence of particles in an interacting system as $N \rightarrow \infty$.

Resultant length

See order parameter.

Sliced Wasserstein distance

A tractable distance between high-dimensional distributions, computed by averaging one-dimensional optimal transport over random projections. Used—and shown to have a finite-sample floor—in the propagation-of-chaos diagnostics.

Sufficient statistic

A function of the data carrying all the information about a parameter. For von Mises it is $\sum_j \cos \theta_j$, i.e. r —which is why an ML concentration estimate would make the collapse test circular.

Vacuum manifold

The set of lowest-energy field configurations. Its topology classifies the defects.

von Mises distribution

The circular analogue of the Gaussian, $\rho(\theta) \propto e^{a \cos \theta}$.

Winding number

The integer count of how many times a circle-valued phase wraps around a loop; non-zero winding obstructs a global real-valued lift.

C Where to look next

The manuscript.

`main.tex` / `main.pdf`. Section 2 for the model, Section 3 for the prediction and protocol, Section 4 for the simulations, Section 5 for the composition and its table of eight edges, Section 6 for the interpretive commitment.

Supplemental Material.

`supplementary.tex` / `supplementary.pdf`. The claim-by-claim map from statements to Lean identifiers is Table S1; the soundness argument, the fold-and-cross proof, the compatibility-observable specification and the full EEG methods are separate sections there.

The Lean development.

`PhysicsOfConsciousness/`, organized by phase. `Chain.lean` is the composition; `Examples.lean` carries every witness and every fencing counterexample, and is the most useful file for seeing what the hypotheses actually demand.

Simulations.

simulations/, with saved compact summaries and the generator that turns them into the numerical macros used by the manuscript, the Supplemental Material and this primer.

References

- Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable electric fields emerge from and allow representational drift. *NeuroImage*, 253, 119058.
- Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7), 450-461.
- Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the Mind. *Oxford University Press*.
- Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a computational theory of conscious processing. *Current Opinion in Neurobiology*, 25, 76-84.
- Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature Reviews Neuroscience*, 11(2), 127-138.
- Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows memory network formation. *Cerebral Cortex*, 33(17), 9877-9890.
- Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling: Electric fields sculpt neural activity and "tune" the brain's infrastructure. *Progress in Neurobiology*, 226, 102465.
- McFadden, J. (2002). The conscious electromagnetic information (cemi) field theory: the hard problem made easy?. *Journal of Consciousness Studies*, 9(8), 45-60.
- Pockett, S. (2012). The electromagnetic field theory of consciousness: a testable hypothesis about the characteristics of conscious as opposed to non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.
- McFadden, J. (2020). Integrating information in the brain's EM field: the cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1), niaa016.
- Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may guide neocortical network activity. *Neuron*, 67(1), 129-143.
- Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous electric field activity: why bother?. *Current Opinion in Neurobiology*, 31, 95-103.
- Pockett, S. (2002). Difficulties with the electromagnetic field theory of consciousness. *Journal of Consciousness Studies*, 9(4), 51-56.
- Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation on brain circuits in rats and humans. *Nature Communications*, 9, 483.
- Verkhatsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physiological Reviews*, 98(1), 239-389.
- Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Correa, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced unconsciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.

- Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J. F., & Juel, B. E. (2025). A repeated awakening study exploring the capacity of complexity measures to capture dreaming during propofol sedation. *Scientific Reports*, 15, 32012.
- Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... & Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult brain. *Science*, 342(6156), 373-377.
- Syková, E., & Nicholson, C. (2008). Diffusion in brain extracellular space. *Physiological Reviews*, 88(4), 1277-1340.
- Barbour, B. (2017). Analysis of claims that the brain extracellular impedance is high and non-resistive. *Biophysical Journal*, 113(7), 1636-1638.
- Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return of Maxwell's demon. *Studies in History and Philosophy of Science Part B: Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.
- Landauer, R. (1961). Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3), 183-191.
- Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal of Physics A: Mathematical and General*, 9(8), 1387-1398.
- Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.
- Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics of prediction. *Physical Review Letters*, 109(12), 120604.
- Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation: The phase-space perspective. *Physical Review Letters*, 98(8), 080602.
- Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L. (2026). Thermodynamic learning. *arXiv preprint*. [arXiv:2609.04732](https://arxiv.org/abs/2609.04732).
- Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-181.
- Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-archive.pitt.edu/115.
- Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen, H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in natural systems. *Nature Reviews Physics*. [doi:10.1038/s42254-026-00971-4](https://doi.org/10.1038/s42254-026-00971-4).
- Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics, vol. 1464, pp. 165–251. Springer. [doi:10.1007/BFb0085169](https://doi.org/10.1007/BFb0085169).
- Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin, P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11), 4657-4662. [doi:10.1523/JNEUROSCI.4509-14.2015](https://doi.org/10.1523/JNEUROSCI.4509-14.2015).
- Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave patterns underlie complex brain dynamics and are related to cognitive processing. *Nature Human Behaviour*, 7(7), 1196-1215. [doi:10.1038/s41562-023-01626-5](https://doi.org/10.1038/s41562-023-01626-5).